A Semantic Mutation Metric for Metamorphic-Relation Adequacy in Scientific Computing Programs

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Metamorphic testing mitigates the test-oracle problem by checking relations among executions, but classical mutation score is defined over syntactic edits and does not say whether a metamorphic-relation set observes declared domain-semantic effects. This paper introduces Semantic Mutation Score (SMS), an MR-relative adequacy metric over an admitted universe of nonequivalent semantic mutants with explicit certificates and a degeneration path back to classical mutation score. We instantiate five semantic operator families on 12 single-output scientific-computing programs and audit the resulting 60-cell design with an SMS-to-MS proof, an AST-normalized syntactic comparison, aligned-versus-cross relation analysis, and boundary and adjoint studies. The semantic pool has 5.14

CCS Concepts: • **Software and its engineering → Software testing and debugging**; *Software verification and validation.*

Additional Key Words and Phrases: metamorphic testing, mutation testing, semantic mutation testing, metamorphic relation adequacy, metamorphic relation assessment, scientific computing

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1 Introduction

1.1 Software-Engineering Motivation and Scope

Metamorphic Testing (MT) addresses the test-oracle problem in scientific computing software: instead of checking outputs against a known-correct reference, MT checks metamorphic relations (MRs), invariants connecting outputs of related inputs. While *semantic mutation testing* has been named since Clark et al. [14] and tooled for C [19] and for UML state-machine behaviour [23], and while MR-adjacent data-semantic mutation [1, 2, 12, 53] and recent LLM semantic-invariance MT [17] extend the mutation target to test data, relations, and task-level semantics, the adequacy of an MR set against *domain-specific code-level faults*, conservation laws, monotonicity, convergence order, trajectory shape, fidelity ordering, has lacked a backward-compatible metric tied to the classical Mutation Score (MS) lineage. Jia and Harman's [2011] MS is defined over syntactic Abstract Syntax Tree (AST) mutations and does not capture these

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domain semantics. Recent Large Language Model (LLM) generated mutants [31, 47] do produce semantically richer mutants, but do not separate the contribution of LLM source diversity from the contribution of MR design.

Throughout this paper we use the term *four representative classes of single-output scientific computing kernels* (numeric, probabilistic, surrogate, and machine-learning), and avoid the ambiguous software-engineering term *paradigm*. The scope of our claims is strictly bounded to single-output float-to-float kernels: each Program Under Test (PUT) is under 2 KB. We do not claim industrial transfer for multi-output software in this paper.

The four thresholds this study pre-registered for itself, operator instantiability (H1), aligned-versus-cross effect size (H2), cross-class consistency (H3), and kill-attribution purity (H4), are each not met. We report those misses plainly and treat them as the diagnostic result: they are boundary-delimiting findings that locate where operator applicability, MR design, and kill attribution break down, not outcomes to be softened. Against that backdrop, an industrial real-defect arm then tests the same semantic distinction developed here on reproduced defects from widely used scientific-computing libraries: mutation kill-rate, semantic stratum alignment, and real-defect detection are related but different constructs. The arm strengthens the semantic-mutant argument at result level; it is not presented as a reusable benchmark artifact. Its cases were admitted only after defect/MR detectability verification and are reported here at result level. Candidate mining, rejected cases, repository tooling, and benchmark release are outside this manuscript and remain contributions of the archived dataset it draws on.

The paper is organised as three studies with a narrow, explicit claim ladder. *Study 1* is the 12-PUT audit, its structural and industrial arms, and the four threshold misses just described; it *delimits* the construct. *Study 2* (Section 6) re-registers those misses as powered successors on a blind-authored 28-PUT grid, run once: three confirm (operators instantiate adequately H1′, the aligned slice dominates the cross slice directionally H2-1′, the direction holds in 3/4 classes H3′), attribution purity H4′ is *not* confirmed (multi-stratum leakage generalises beyond Study 1's CF/TF diagnosis once the grid widens), and a cross-vendor question is gated not-run under the same-vendor harness. *Study 3* (Section 7) re-registers attribution alone on fresh mutants with the repaired screen verified live, and splits it into a strict and a graded verdict: single-stratum attribution is *confirmed* where coupling is absent or screenable (CE, HP, CF), while graded declared-stratum attribution is *refuted* on the rich surrogate and ML classes. Study 1 delimits, Study 2 confirms directionally (without upgrading the medium effect into a large-effect claim), and Study 3 bounds the attribution boundary on both sides.

The intended reader is a software-testing researcher or practitioner who needs to decide whether a set of MRs is adequate for a declared semantic risk, not a domain scientist seeking a new numerical solver or a tool user seeking another LLM-mutant generator. The novelty is therefore the adequacy denominator and certificate structure: SMS asks which admitted semantic-effect fibers are observed by an MR set, while preserving the classical mutation-score form. The practical impact is diagnostic rather than deployment-ready automation: the metric exposes when MR design, operator applicability, or kill attribution prevents a semantic adequacy claim from being trusted.

1.2 Three-layer methodological framework

We organise the contribution as a three-layer framework around domain-semantic mutation operators.

- **Layer 1, Definitional (Section 4.2).** A mutation is *semantic* if it satisfies at least one of three necessary conditions: (a) it crosses a function-call or module-import boundary, (b) it depends on domain knowledge for legality, or (c) it changes the algorithmic class. The five operator families, Conservation Erosion (CE, also written mut_C), Operator Substitution (OS, mut_M), Hyperparameter (HP, mut_G), Trajectory Flip (TF, mut_T),

Table 1. Evidence-to-claim map for the empirical argument.

Claim supported	Evidence used	Claim not made
SMS is backward-compatible with classical MS	Formal SMS-to-MS degeneration under the syntactic limit	SMS replaces classical mutation score in ordinary syntactic settings
Semantic mutants occupy a different admitted universe	5.14% AST-normalised overlap with default first-order syntactic mutants; HP, SI, and TF are absent from the default first-order pool	Default syntactic mutation tools can never generate higher-order semantic mutants
MR alignment is visible but bounded	Aligned mean SMS 0.213 vs cross mean SMS 0.077 (MP5-held pool); Cliff's δ is positive but below the pre-registered large-effect threshold	The current 12-PUT audit proves a large universal aligned-vs-cross effect
SMS exposes adequacy boundaries	H1–H4 failures, zero-mass cells, root-cause labels, and strong-boundary false-positive / false-negative cases	Failed thresholds are treated as empirical success criteria
Real-defect behaviour is distinct from aggregate kill-rate	The industrial real-defect arm: a pre-registered four-group mutation comparison plus a per-defect detection face on reproduced library defects	The manuscript contributes a reusable real-defect benchmark

and Structural Injection (SI, mut_F), specialise these conditions across the four PUT classes. We call these five sets *operator families* throughout and reserve *meta-pattern* for the five MR strata (Section 3.3.2); the two are distinct five-element sets placed in one-to-one alignment, so we avoid the label "meta-operator" to keep them apart.

- **Layer 2, Operational (Section 3.3.3, Section 4.10).** An equivalence judgement E1 ∧ E2, where E1 is Automated Verification Pipeline (AVP) coherence and E2 is output equivalence on K_eq = 1000 samples, gives the conservative instantiation of the Layer-1 conditions. The trade-off against E1-alone and E2-alone variants is in Appendix A.3.
- **Layer 3, Applied (Section 4.5).** AST-normalised empirical traceability across all 12 PUTs: comparing 292 v4 mutants against 1,250 cosmic-ray syntactic mutants, we show empirically that the v4 pool is not a subset of the syntactic-mutant pool.

The 60-cell empirical audit reported in Section 5 stress-tests this backbone. SMS is positioned as a strict generalisation of classical MS: under the degenerate limit $L = L_{\text{equiv}} \wedge L_{\text{killed}} \wedge L_{\text{mut}}$ formalised in Section 3.3.6, SMS reduces almost everywhere to MS. Layer 1 and Layer 3 correspond to the two leading research questions, RQ1 (constructibility and backward compatibility) and RQ2 (syntactic unreachability); the 60-cell demonstration answers RQ3–RQ5.

The empirical evidence has four roles, read in sequence: the 12-PUT audit is the main stress test of the SMS construction; the AST audit tests whether the semantic-mutant space is reachable by default first-order syntactic mutation; the boundary and adjoint arms map when strong MRs do and do not detect semantic effects; and the industrial real-defect arm checks whether the construct distinctions remain visible on verified MR-detectable defects outside synthetic mutants and author-written kernels.

This table also marks the paper's non-claims. We do not claim that SMS is universally superior to classical MS, that pattern-derived MRs dominate all generic relations, that higher-order mutation has been refuted, that the industrial real-defect arm is a benchmark contribution, or that the reported SMS values define industrial acceptance thresholds.

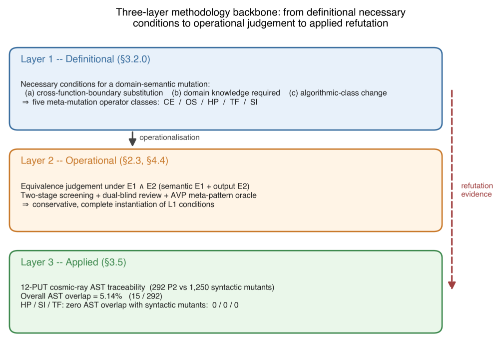

Fig. 1. Three-layer methodological framework: Layer 1 (definitional necessary conditions), Layer 2 (operational $E_1 \wedge E_2$ judgement), Layer 3 (applied AST-normalised traceability across 12 PUTs).

2 Related Work

Seven lines of prior work bracket the contribution of this paper.

General MT foundations and surveys define the oracle-problem setting and the role of metamorphic relations in paired-test reasoning [13, 41, 46]. Because our empirical subject class is scientific computing, we also take the scientific-software testing survey by Kanewala and Bieman [37] as the nearest domain background.

(a) Classical mutation testing. Jia and Harman [35] and Papadakis et al. [43] survey syntactic mutation, and the Coupling Effect Hypothesis (CPH) underpins MS as a fault proxy [6, 22, 36]. We argue in Section 4.5 that CPH holds within syntactic mutation but does not automatically extend across the syntactic-versus-domain-semantic boundary, with the 5.14% AST overlap (HP, SI, TF at 0/0/0) as the empirical witness. The correctness-versus-test-evidence concern predates modern mutation score [11]; SMS inherits it but changes the adequacy object from a test set to an MR set over admitted semantic mutants.

(b) Higher-order mutation. Jia and Harman [34] and Kintis et al. [38] study compositions of first-order syntactic mutants. We explicitly exclude any equivalence claim about Higher-Order Mutation (HOM) and list HOM equivalence as a residual threat (Appendix F.2).

(c) LLM-generated mutants. Pre-trained-language-model mutation begins with μBERT [20]; Tip et al. [47] introduce LLMorpheus (single-LLM JavaScript mutants), Humbatova et al. [31] DeepCrime (deep-learning real-fault mutation), and Dakhel et al. [18] extend LLM-driven mutation to Java test generation. To our knowledge no prior LLM-mutant work isolates *LLM source diversity* from *MR design* in the effect size; the ablation in Section 4.9 closes this gap. On equivalence, Delgado-Pérez and Chicano [21] emphasise the `equiv` term, which we extend to a semantic-class equivalence E1 ∧ E2 (Section 3.3.3); Zhang et al. [52] validate MT in class-integration ordering; and Meta's ACH [25] deploys mutation-guided LLM test generation on general software, whereas we measure MR adequacy on scientific kernels.

(d) Semantic mutation operators across testing communities. Semantic mutation predates the LLM-mutant lineage: Clark et al. [14] introduce *semantic mutation testing* distinct from syntactic mutation, Dan and Hierons [19] tool it for C, and Derezińska and Zaremba [23] apply *semantic mutation operators* to UML state machines by perturbing semantic-variation-point interpretations. Their object is the language interpretation a developer may misread; ours is a certified violation of a declared domain invariant in code. Within MT, ai Sun et al. [1, 2] propose *data mutation*

Table 2. Closest denominator-oriented comparators to SMS. The comparison is by adequacy object and denominator semantics, not by whether a paper uses the phrase metamorphic-testing adequacy.

Work	Denominator or counted object	Relation to SMS
[26, 42]	MR obligations, MR coverage, and source-test obligations	Closest MR/MT adequacy comparators, but property/coverage denominators rather than semantic mutant classes
[8]	Differentially executed code elements in paired MT runs	Execution-side coverage comparator, not MR-set semantic adequacy
[50]	MR feature/diversity space	Boundary surrogate for MR-set quality, not an adequacy fraction over faults or semantic classes
[30, 51]	MR validity, coverage, and classical mutation-score signals	Boundary surrogate for cost-effective MR choice, not a denominator-bearing adequacy metric
[10]	Semantic obligations under a specification	Non-MT semantic-coverage analogue; test-suite side rather than MR-set side

operators and the μMT methodology, Zhu et al. [53] formalise *datamorphic testing* (separating datamorphisms from metamorphisms), Chan and Keung [12] extend generic data mutation to defect-prediction validation, and Curtò and Zarzà [17] formalise *semantic invariance* in LLM MT. These establish that mutation can target data, relation, task, and behavioural semantics. **The contribution of this paper is the first to formalise a domain-specific Semantic Mutation Score (SMS) for MR adequacy in scientific computing kernels, with a proven backward-compatible reduction to classical MS (Theorem 3.1) and a code-level operator framework (Conservation Erosion, Operator Substitution, Hyperparameter, Trajectory Flip, Structural Injection) targeting kernel-specific faults, and to delineate the strong boundary of metamorphic-relation grounding, a strong/weak MR distinction whose ε_{tol}-governed duality (Theorem 3.5, Proposition 3.6) is mapped empirically on real and textbook programs spanning the meta-patterns (Section 5.6), complemented by a cross-library adjoint extension arm (Section 4.7, Section 5.7) that supports the duality forward on third-party linear operators with real fixed defects.**

(e) MR adequacy and coverage. Recent MT adequacy work, separated by its *counting object*: Xiang et al. [49] build composite MRs but evaluate with the classical MuJava syntactic-mutant denominator; Fu et al. [26] define MT adequacy over MR obligations and source inputs; Liu et al. [42] integrate MR coverage; and Ba et al. [8] define Metamorphic Coverage over differentially executed code elements, all denominators being obligations, source-tests, or execution coverage rather than semantic mutant classes. Xie et al. [50] quantify MR diversity (MUT), and Huang et al. [30], Zhang et al. [51] rank/select MRs by validity, coverage, or mutation-score, all boundary surrogates rather than an adequacy fraction over enumerated semantic fault classes. Blwi et al. [10] give a non-MT semantic-coverage analogue on the test-suite side. None makes the denominator the admitted nonequivalent *domain-semantic code mutants* for scientific kernels or establishes compatibility with the classical MS denominator; SMS differs by replacing the denominator against which an MR set is judged, not by proposing another coverage metric.

(f) Property-relative and numerical-specification mutation. Recent property-aware mutation work moves the adequacy question closer to requirements and invariants: Bartocci et al. [9] make mutant relevance and meaningful killing relative to a tested property, and Jeangoudoux et al. [32] generate tests for numerical specifications through interval-constraint reasoning over floating-point accuracy and robustness requirements. These works are closer to SMS than generic AST mutation because they reject property-irrelevant mutants as an adequacy denominator. They still differ from the present paper in three respects: the adequacy object here is an MR set rather than a single formal requirement, the denominator is a five-class domain-semantic mutant space for scientific kernels, and the metric is constructed as a backward-compatible generalisation of MS.

(g) Semantic foundations and the certificate gap. Abstract interpretation supplies the standard machinery for relating concrete semantics to abstract domains and invariant families [15], and for designing program transformations by their effect on an abstract semantics [16]. Approximate program transformation semantics gives a second, quantitative route: local error expressions and partial-metric semantics can certify that a transformation lies within a bounded semantic-effect envelope [27, 48]. Refinement- and UTP-based mutation testing lift mutation to contracts and specifications [3, 4], while usage-aware differential dynamic logic tracks admissible proof-level mutations under soundness conditions [24]. These foundations supply much of the machinery needed for certificate-bearing semantic-effect classes, but they are not MR adequacy metrics for scientific-computing kernels. Conversely, SMS provides the MR adequacy denominator and empirical audit, but its certification layer is empirical and structural rather than machine-checkable proof-carrying.

Closest comparison. The closest adjacent work is the Min-MR-Complete formulation [40], which uses the same semantic-mutation family as the fault model for MR-subset completeness and identifies when class-level reasoning collapses to a mutant-level set-cover problem. The present paper differs in role: it defines the SMS denominator, the E1 $\wedge$ E2 admission judgement, the AST-normalised syntactic-unreachability audit, and the empirical same-source/cross-source ablation. Min-MR-Complete optimises over an admitted mutant–draw coverage universe; this paper constructs and audits that universe. Taken together, the two papers come closest to semantic-effect-class denominators for MR adequacy, but neither yet attaches a per-mutant machine-checkable certificate showing that a syntactic edit belongs to a declared abstract semantic-effect class. Throughout this paper, a certificate is a recorded, budget-stamped piece of test evidence, not a proof object (Limitation 9); the undecidability result is precisely why nothing stronger is available.

The SMS metric is conceptually complementary to the code-verification scope of ASME V&V 20-2009 §3, which targets numerical-solver correctness. SMS targets MR fault-detection adequacy; we make no normative compliance claim (Appendix E.3 records this as a long-term aspiration only).

3 Problem Formulation and Semantic Mutation Model

This section turns the problem into auditable research questions before the metric: scope and novelty are stated as RQ1–RQ2, while RQ3–RQ5 test whether one empirical instantiation is usable, discriminative, and consistent enough to support the construct.

3.1 Research Questions

The questions run from the theoretical foundation (RQ1), through the structural separation (RQ2), to the 60-cell demonstration (RQ3–RQ5). RQ1 is answered by construction and proof (Sections 3.3.7–3.3.9: Theorem 3.3, Proposition 3.4, Theorem 3.5, and the degeneration of Section 3.3.6); RQ2 by the AST audit (Section 4.5); RQ3–RQ5 by the pre-registered pipeline (Section 5), mirroring the three-layer framework (Layer 1 carries RQ1, Layer 3 RQ2).

- **RQ1 (Theory of semantic mutation: constructibility, undecidability, duality, backward compatibility).** Can semantic mutation be formalised as a fiber decomposition of a single effect map over admitted nonequivalent *domain-semantic code mutants* for scientific-computing kernels, such that (a) stratum membership is undecidable and therefore certificate-based (Theorem 3.3), (b) the induced adequacy metric SMS degenerates almost everywhere to the classical Mutation Score (Proposition 3.4; Theorem 3.1), and (c) strong metamorphic relations detect only the semantically active mutants within a tolerance-bounded regime, while outside it the duality degrades along an ε_{tol}-governed strong boundary into false positives (weak MRs, where a correct discrete program violates the invariant) and false negatives (surviving non-equivalent mutants under coincidental satisfaction) (Theorem 3.5 and Proposition 3.6; the boundary is mapped empirically under RQ4)?

- **RQ2 (Syntactic unreachability).** To what extent is the domain-semantic mutant space induced by the five operators reachable by default-configuration first-order syntactic mutation tools at the AST-normalised level?
- **RQ3 (Instantiability).** Do the five semantic mutation operators instantiate a usable nonequivalent semantic-mutant denominator across the 60 PUT-MP-operator cells, as reflected by inst_rate, equiv_rate, C1_share, and survive_rate?
- **RQ4 (Discrimination, LLM-source decoupling, the strong boundary, and industrial construct separation).** Does SMS discriminate aligned (j=k) from cross (j≠k) slices; does cross-source pooling under an identical prompt move the effect size or only the kill-set quality; where does the strong boundary of Proposition 3.6 lie empirically (where does a strong MR kill the genuine fault and where does the duality fail as a weak-MR false positive or a surviving non-equivalent mutant); and does the construct separation among kill-rate, semantic alignment, and real-defect detection remain visible on an industrial arm of reproduced, MR-detectable library defects?
- **RQ5 (Cross-class consistency).** Does the aligned-versus-cross SMS pattern generalise across four representative classes of single-output scientific-computing kernels? (The relationship between SMS and the coverage-style surrogate Pattern Coverage is reported descriptively at n = 12 as a hypothesis-generating observation, not a confirmatory test.)

3.2 Hypotheses

The empirical questions RQ3–RQ5 carry pre-registered hypotheses; RQ1 is answered by Theorem 3.1 and RQ2 by the Section 4.5 structural audit, neither of which is a statistical hypothesis. The hypothesis definitions, primary-MR convention, and statistical reporting hierarchy are frozen in `docs/experiment_documentation/EXPERIMENT_DESIGN.md` at commit `0f2509527f346f9433c3cf90959bb07d80601a23` and in the Zenodo replication package 10.5281/zenodo.20250664.

- **H1 (RQ3).** At least 4 of the 5 operators produce ≥ 5 non-equivalent mutants on at least 9 of the 12 PUTs.
- **H2 (RQ4; pre-registered as an exploratory large-effect target).** The aligned-SMS to cross-SMS odds ratio is ≥ 3.0, and Cliff's δ is ≥ 0.474. The threshold was fixed before data collection, but the design is underpowered at this boundary, so its verdict is read as a point-estimate result (stipulated-alternative power in Section 5.4), not a confirmatory claim of the paper.
- **H3 (RQ5).** A within-class sign test gives 4/4 across the 4 classes, and CV(ΔSMS) < 0.5.
- **H4 (RQ4).** Mean suspect_share is ≤ 0.20 across the 60 cells.

What would count against the construct (stated post hoc). The pre-registered thresholds stress one instantiation and do not exhaust falsifiability. The construct itself would be disconfirmed by: (i) the degeneration path failing empirically, with SMS and MS disagreeing on the same pools in the syntactic limit; (ii) the aligned-versus-cross direction reversing systematically across designs; (iii) the industrial arm showing the generic baselines' kill sets nesting the pattern-derived kill sets, leaving no construct separation; or (iv) certificates failing audit, with declared strata not matching observed effects. Criteria (ii)–(iv) were observed not to occur; criterion (i) was not directly tested and is open to replication. We state these criteria post hoc, after the verdicts, and label them accordingly.

The 12 PUTs are deliberately *Numerical Recipes*–style compact kernels: they provide a verifiable minimum working example for the three-layer backbone, while industrial-scale transfer is left to future work under the representativeness threat.

3.3 Formal Model and Semantic Mutation Score

3.3.1 Vocabulary and Notation. We use a small vocabulary throughout the method. The terms are declared here before they are used in the SMS construction.

Program Under Test (PUT). A PUT is the scientific-computing kernel being assessed. We write the i-th PUT as S_i.

Metamorphic Relation (MR). An MR is a property that relates the outputs of related inputs. The MR family used for PUT S_i and meta-pattern k is written $\mathrm{MR}_{i,k}$. This $\mathrm{MR}_{i,k}$ is a per-PUT, per-meta-pattern *instance* of a meta-pattern, that is, the concrete relation set exercised on one subject at one stratum; in the companion framework NOETHER [7] the coarser object $\mathrm{MR}(\mathcal{A}_P)$ is the algebra-induced MR space of a whole program family, and a MetaPattern is an equivalence class over it, so our $\mathrm{MR}_{i,k}$ sits one granularity below NOETHER's MR class.

Meta-pattern (MP). An MP is a semantic stratum of MR behaviour, such as conservation, monotonicity, convergence, trajectory structure, or fidelity order. The index k denotes the MP being tested.

The five meta-patterns are the instance, on the scientific-computing program family, of the canonical MetaPattern taxonomy of NOETHER [7], which derives MetaPatterns as closure-guaranteed equivalence classes over an operator algebra. We adopt NOETHER's names as the primary narrative vocabulary and record the correspondence to our registered labels once, here: MP1 conservation = m_{inv} (symmetry block G), MP2 monotonicity = m_{mono} (order block $O_{\leq}$), MP3 convergence order = m_{conv} (limit block $\mathcal{L}^*$), MP4 trajectory determinism = m_{dyn} (qualitative-dynamics block $\mathcal{D}^*$), and MP5 partial-order consistency = m_{cmp} (method-comparison block $\mathcal{E}^*$); the adjoint extension ψ_6 is m_{adj} (self-adjoint block T^*). The registered labels MP1–MP5 are retained throughout for every pre-registration, dataset-key, and provenance reference, because the pre-registered protocol was authored under those labels and only under those labels; NOETHER's m_{xxx} names carry the semantic narrative and never stand in for a registration. Our framework does not exercise NOETHER's time-reversal (m_{rev}) or relational-equivalence (m_{rel}) blocks.

Semantic operator. A semantic operator is a mutation family mut_j designed to perturb a declared MP. The operator index j identifies the intended semantic effect.

Semantic mutant. A semantic mutant s' is a finite syntactic edit of S_i that is admitted by semantic operator mut_j and is certified against the declared MR family. Formally,

$$s' \in \mathrm{mut}_j(S_i).$$

The word semantic therefore names the certification target, not a second editing mechanism.

Semantic-effect fiber. For an MR family $\mathrm{MR}_{i,k}$, the semantic-effect fiber is the subset of admitted semantic mutants whose declared effect is observable by that MR family:

$$\mathcal{F}(\mathrm{MR}_{i,k}) = \{s' \in \mathrm{mut}_j(S_i) \,:\, \mathrm{killed}(s', \mathrm{MR}_{i,k})\}.$$

Different MR families can observe different fibers. SMS therefore measures adequacy relative to a declared stratum; it is not a total order over all possible MR families.

Table 3. Operator signatures.

Operator	Failure semantics	Aligned MP
mut_C	Conservation-breaking	MP_1 Conservation
mut_M	Monotonicity-breaking	MP_2 Monotonicity
mut_G	Convergence-breaking	MP_3 Convergence
mut_T	Trajectory-distorting	MP_4 Trajectory
mut_F	Fidelity-order-breaking	MP_5 Partial-order

Equivalent, killed, and surviving mutants. A mutant is equivalent for (i, k, j) when the equivalence judgement of Section 3.3.3 admits it into $\text{equiv}_{i,k,j}$. A non-equivalent mutant is killed when at least one MR in $\text{MR}_{i,k}$ passes on S_i and fails on s'. A non-equivalent mutant that is not killed is surviving. Thus the generated mutant set decomposes as:

$$\text{mut}_j(S_i) = \text{equiv}_{i,k,j} \cup \text{killed}_{i,k,j} \cup \text{survive}_{i,k,j} \quad \text{(disjoint)}$$

$$\text{SMS}_{i,k,j} := \frac{|\text{killed}_{i,k,j}|}{|\text{mut}_j(S_i)| - |\text{equiv}_{i,k,j}|} \in [0,\ 1]$$

Semantic Mutation Score (SMS). SMS preserves the classical Mutation Score (MS) ratio while replacing the internal definitions of mutant generation, equivalence, and killing with MR-relative semantic definitions. The classical concepts of mutation operator, mutant, equivalent mutant, killed mutant, surviving mutant, and MS are inherited from Jia and Harman [35] and Ammann and Offutt [5].

The complete notation table is in Appendix A.1; index sets and tolerance parameters ($\varepsilon_{\text{eq}}, \varepsilon^k_{\text{AVP}}, K_{\text{eq}} = 1000, N = 20$) follow that table.

3.3.2 Semantic-Operator Signatures. Operator family and alignment:

$$\text{MUT} = \{\text{mut}_C,\ \text{mut}_M,\ \text{mut}_G,\ \text{mut}_T,\ \text{mut}_F\} \quad \text{(open, extensible)}$$

$$\text{mut}_j\ :\ \text{Programs} \to 2^{\text{Programs}}$$

$$\text{align}(j) = j \quad \text{(design choice, not theorem)}$$

Per-PUT specialisation tables (mut_C/M/G/T/F across 12 PUTs and 4 classes) are deferred to **Appendix B.2**.

3.3.3 Equivalence Judgement. The equivalence judgement is the executable instantiation of the Layer-1 necessary conditions (Section 4.2). For each candidate mutant `s'`:

- **E1 (AVP-coherent):** ∀ mr ∈ $\text{MR}_{i,k}$: $\text{AVP}(S_i, mr) = \text{AVP}(s', mr)$.
- **E2 (Output-equivalent):** ∀ x in $K_{\text{eq}} = 1000$ samples ∼ D_S: $\|S_i(x) - s'(x)\| \le \varepsilon_{\text{eq}}$.

E1 ∧ E2 is the conservative complete instantiation: false-equiv (a truly non-equivalent mutant admitted as equivalent) requires *both* AVP coherence and numerical agreement on K_eq samples to hold despite the non-equivalence (both mis-pass simultaneously), which is rare; false-non-equiv requires only one of E1, E2 to fail for a truly equivalent mutant, biasing SMS slightly high. The trade-off table against E1-alone and E2-alone is in Appendix A.3. Under the Section 3.3.6 degenerate limit L, E1 ∧ E2 reduces almost everywhere to classical bitwise equivalence (Lemma G.1, supplementary Appendix G).

Semantic certificate record. For each admitted mutant, the empirical certificate is a structured record rather than a proof-carrying object. The record contains the PUT identifier, operator identifier, intended meta-pattern stratum, edit class, targeted semantic invariant, E1 AVP-coherence result, E2 sampling budget and tolerance, killed/survived vector over the relevant MRs, LRCA labels for killed outcomes, generation source and reviewer status, and a trace path to the reproduced artifact. A mutant enters the SMS denominator only after the record passes the Layer-1 semantic-condition check and the E1 ∧ E2 equivalence judgement. LRCA labels remain diagnostic annotations on the certificate; they do not filter the denominator or redefine SMS.

The killed determination preserves the classical OR-aggregation:

$$\text{killed}(s', \text{MR}_{i,k}) \iff \exists\, mr \in \text{MR}_{i,k} \;:\; \text{AVP}(S_i, mr) = \text{pass} \wedge \text{AVP}(s', mr) = \text{fail}.$$

3.3.4 Root-Cause Diagnostic Layer. Likely Root-Cause Analysis (LRCA) annotates every killed mutant with one of five diagnostic labels: C1 likely semantic failure, C2 numerical-tolerance perturbation, C3 out-of-distribution (OOD) trip, C4 statistical-assumption violation, and C5 mutator artefact. The diagnostic procedure is a three-layer decision tree. Layer 1 checks tolerance robustness over N = 20 repeats with a fail-ratio cutoff of 0.80. Layer 2 triages OOD on the surrogate and machine-learning classes. Layer 3 checks statistical-assumption baselines on the probabilistic and machine-learning classes using Wilcoxon and Dynamic Time Warping. A final pass rechecks for mutator artefacts. Multiple labels are resolved by priority C5 > C4 > C3 > C2 > C1.

The output quantities are `C1_share` (share of C1 among killed mutants) and `suspect_share := 1 − C1_share`. **LRCA does not modify the SMS formula**: the killed set is never filtered by suspect status. LRCA is a diagnostic annotation that indicates whether a given kill is more consistent with a semantic-failure detection, tolerance sensitivity, OOD behaviour, statistical-assumption failure, or mutator artefact. Appendix A.2 gives the full decision tree, the 9-grid threshold calibration, and the engineering rationale for the priority ordering.

3.3.5 Backward-compatibility declaration. The seven core concepts (PUT, mutation operator, mutant, equivalent mutant, killed mutant, surviving mutant, mutation score) and the SMS formula `|killed| / (|mut| − |equiv|)` are aligned item-for-item with Jia and Harman [35]. We extend only the internal definitions: `mut` shifts from syntactic to domain-semantic operators, `equiv` shifts from bitwise behavioural equivalence to the semantic-class equivalence E1 ∧ E2, and `killed` shifts from an equality oracle to a meta-pattern AVP. We introduce no new formula terms and no new state classifications.

Under the degenerate limit L defined in Section 3.3.6, SMS reduces almost everywhere to classical syntactic MS, equivalent mutants degenerate to classical behavioural equivalence, killed mutants degenerate to bitwise difference detection, and the LRCA layer trivialises, C2 through C5 cannot fire when L1 ∧ L2 ∧ L3 hold. Any SMS-based empirical conclusion is therefore structurally consistent with the existing mutation-testing literature in the classical syntactic-mutation regime, so there is no metric-level semantic fragmentation. The skeleton of eleven concepts, the three-state decomposition, the SMS formula, and the AVP interface are fixed; the contents of MUT, MP, C, and cls remain open to future extension.

3.3.6 Degeneration to Classical Mutation Score. The Section 3.3.5 backward-compatibility claim is formalised as a degeneration theorem. The full notation cross-reference is **Appendix G.1**; the joint conditions are in **Appendix G.2**, the lemma proofs in **Appendix G.3**, and the full theorem proof in **Appendix G.4**. We state only the main theorem and corollary here; the three construction lemmas are stated and proved in the supplementary material as Lemma G.1–G.3.

Definition (Degenerate limit). $L = L_{\text{equiv}} \wedge L_{\text{killed}} \wedge L_{\text{mut}}$, where each joint condition is a pair of paired axes acting on one layer of the SMS formula:

- $L_{\text{equiv}} = L1 \wedge L2$: $\varepsilon_{\text{eq}} \to 0 \wedge K_{\text{eq}} \to \infty$ (controls equiv layer; Lemma G.1).
- $L_{\text{killed}} = L3 \wedge L4$: $\varepsilon_{\text{AVP}} \to 0 \wedge$ MP set = $\{\text{MP}_{\text{eq}}\}$ (controls killed layer; Lemma G.2).
- $L_{\text{mut}} = L5 \wedge L6$: mut_j switches to rule-based syntactic operators (Mothra-style, or modern MutMut/PIT-style) $\wedge$ PUT class $\subseteq$ imperative deterministic programs (controls mut layer; Lemma G.3).

Theorem 3.1 (SMS → MS degeneration). *In the degenerate limit L, **almost everywhere** with respect to the input distribution D_S,*

$$\text{SMS}_{i,k,j} \xrightarrow{L} \text{MS}_{i,j} = \frac{|\text{killed}_{i,j}^{\text{classic}}|}{|\text{mut}_j^{\text{syntax}}(S_i)| - |\text{equiv}_{i,j}^{\text{classic}}|}$$

where the right-hand side is the classical Mutation Score of Jia and Harman [35], $\text{killed}^{\text{classic}}$ *is the difference-detection set,* $\text{equiv}^{\text{classic}}$ *is the behavioural-equivalence set, and* $\text{mut}^{\text{syntax}}$ *is the syntactic-mutant set.*

Proof sketch. By Lemma G.1, equiv_{i,k,j} → equiv_{i,j}^classic almost everywhere with respect to D_S; the measure-zero qualifier accommodates floating-point pathological points and Not-a-Number propagation. By Lemma G.2, killed_{i,k,j} → killed_{i,j}^classic, and the MP index k degenerates because L4 collapses MR_{i,k} to {MP_eq}. By Lemma G.3, mut_j(S_i) → mut_j^syntax(S_i). Substituting into the SMS formula gives MS_{i,j}. Full lemma proofs are in supplementary Appendix G.3 and the full theorem proof in supplementary Appendix G.4. □

Under the stated substitutions the equality is definitional once the three layers coincide; the almost-everywhere qualifier concerns only the bounded E2 classifier on floating-point pathological inputs. We therefore read Theorem 3.1 as a backward-compatibility characterisation of SMS, not as an independent mathematical contribution.

Corollary 3.2 (LRCA trivialisation). *Under L, the likely-root-cause inventory C = {C1, ..., C5} degenerates to {C1}, so suspect_share → 0 and SMS becomes a single-layer metric consistent with the engineering attribution structure of Jia and Harman [35].*

Empirical consistency. Theorem 3.1 and Corollary 3.2 jointly show that the SMS-based empirical conclusions (Section 5 Cliff's delta, Friedman chi^2, Spearman rho) are structurally consistent with existing mutation-testing literature in the classical syntactic-mutation regime, and **do not** constitute metric-level semantic fragmentation.

3.3.7 Semantic-Mutation Theory Setting. The preceding subsections fix the SMS metric and its degeneration to MS; we now give the theory it instantiates. Semantic mutation is *intensionally semantic, extensionally syntactic*: every mutation is mechanically a syntactic act (semantics changes only through syntax), and what makes a mutant *semantic* is where its defining condition lives (a specified invariant violation at the semantic layer) and how its membership is established (by certificate, because the condition is undecidable).

Fix a grammar G and the set $\text{Prog}(G)$ of well-formed programs with denotational semantics $[\![\cdot]\!] : \text{Prog}(G) \to (\mathcal{X} \rightharpoonup \mathcal{Y})$. Let $\alpha : \mathcal{Y} \to \mathcal{Y}_\alpha$ be the committed semantic abstraction (the observable that the AVP of Section 3.3.3 reads), and write $P \equiv_\alpha P'$ when $\alpha \circ [\![P]\!] = \alpha \circ [\![P']\!]$ pointwise on the admitted input domain $\mathcal{X}_{\text{adm}}$. This $\equiv_\alpha$ is exactly the equivalent-mutant relation equiv of Section 3.3.1: the E2 component of the E1∧E2 judgement (Section 3.3.3) is its bounded decision procedure on K_{eq} samples.

The new ingredient is a finite *semantic invariant family* $I = \{\psi_1, \dots, \psi_5\}$, where each ψ is a predicate over the observable behaviour $\alpha \circ [\![P]\!]$. The five invariants are exactly the properties the meta-patterns of Section 3.3.2 target: ψ_1 conservation (m_{inv}), ψ_2 monotonicity (m_{mono}), ψ_3 convergence order (m_{conv}), ψ_4 trajectory determinism (m_{dyn}), ψ_5 partial-order consistency (m_{cmp}), so ψ_k is the invariant whose violation MP_k detects and whose breakage mut_j (at $j = k$) targets; each ψ_k is thus the invariant defining the NOETHER MetaPattern named in parentheses [7]. We write $[\![P]\!] \models \psi$ when P's observable behaviour satisfies ψ.

The family is open (the extensibility of I noted in Section 3.3.9): a sixth invariant ψ_6, *adjoint consistency* (m_{adj}, the self-adjoint block T^* in NOETHER [7]), $\langle Ax, y\rangle = \langle x, A^{\text{H}}y\rangle$ for linear-operator kernels, instantiates the same construction. Its MR is strong (its violation set is closed under $\equiv_\alpha$), so by Theorem 3.5 it detects only the semantically active mutants of the adjoint code path. Because adjoint consistency is meaningful only for linear operators , not for the chaotic, stochastic, and nonlinear machine-learning kernels that make up most of the 12 PUTs, ψ_6 is *not* folded into the pre-registered 60-cell design (doing so would leave the invariant undefined on ten of the twelve subjects); it is instead demonstrated as a separate confirmatory *adjoint extension arm* on dedicated third-party linear-operator subjects (Section 4.7, Section 5.7), leaving the pre-registered hypotheses H1–H4 untouched.

3.3.8 *Semantic-Mutant Conditions.*

The following certified definition is the formal counterpart of the informal Layer-1 necessary conditions of Section 4.2.

Definition (semantic mutant). Given (P, ψ) with $[\![P]\!] \models \psi$, a program P' is a *semantic mutant of P at stratum ψ* iff: (S1) *realizability*, $P' = e(P)$ for a finite AST edit e; (S2) *baseline*, $[\![P]\!] \models \psi$; (S3) *violation with witness*, $[\![P']\!] \not\models \psi$, witnessed by some $x \in \mathcal{X}_{\text{adm}}$ observable through α; (S4) *latency*, P' compiles, terminates on $\mathcal{X}_{\text{adm}}$, and is type-correct, so the defect is observable only at the semantic layer, never by a crash oracle; (S5) *stratum purity* (required where stratum labels feed downstream), $[\![P']\!] \models \psi'$ for all $\psi' \in I \setminus \{\psi\}$.

A semantic mutation class is therefore defined by which invariant must break and how (S2–S5) and realized by whichever syntactic edits achieve it (S1). The operators $\text{mut}_C, \dots, \text{mut}_F$ of Section 3.3.2 are the constructive templates for strata $\psi_1, \dots, \psi_5$.

Definition (latency window). For a violation-magnitude parameter ε of an edit template, the *latency window* is the interval $(\varepsilon_{\text{tol}}, \varepsilon_{\text{crash}})$: above the tolerance at which the ψ-checker witnesses the violation (S3), below the threshold at which the container crashes (S4). An empty window certifies non-realizability of the template at that site.

3.3.9 *Effect map, fibers, and three theorems.*

Definition (effect map and fibers). For fixed P, the *effect map* σ sends an applicable edit e to the semantic-effect class of $P_e = e(P)$, valued in

$$\{\ \equiv_\alpha,\ \text{ill-formed},\ \psi_1\text{-viol}, \dots, \psi_5\text{-viol},\ \text{active-off-taxonomy}\ \},$$

where active-off-taxonomy means $P_e \not\equiv_\alpha P$ yet no $\psi \in I$ flips from satisfied to violated. The *fiber* of a class is its σ-preimage. P_e is *semantically active* iff $P_e \not\equiv_\alpha P$; the strict syntactic-only mutants are exactly the inactive fiber $\sigma^{-1}(\equiv_\alpha)$. σ is single-valued only on the *S5-pure sub-domain* $\{e : |\{\psi \in I : [\![P_e]\!] \not\models \psi\}| \leq 1\}$; on the corpus this sub-domain covers 263/292 (90.1%) of admitted mutants (the S5 audit of Section 5.9). On the residual 9.9%, σ is a relation rather than a function, and those edits are carried explicitly as multi-stratum and excluded from the pure-fiber reading.

Theorem 3.3 (undecidability of semantic-mutant membership). *For Turing-complete* $\mathrm{Prog}(G)$ *and non-trivial* ψ*, deciding whether an edit* e *is a semantic mutant at stratum* ψ *(conditions S2–S4) is undecidable.*

The proof (Rice's theorem for S3, halting reduction for the S4 termination clause) is in supplementary Appendix K.2. The theorem is a routine consequence of Rice's theorem, stated to fix the formal reason admission is gate-shaped rather than a decision procedure, not as a novel result.

Proposition 3.4 (fiber characterization of classical practice). *(i) Every semantic mutant arises from some syntactic edit; the stratum-*ψ *class equals* $\sigma^{-1}(\psi\text{-viol})$ *intersected with S4. (ii) The classical equivalent-mutant problem is exactly the degenerate fiber* $\sigma^{-1}(\equiv_\alpha)$ *of unconstrained syntactic sampling. (iii) Classical mutation testing and semantic mutation are the same effect map under two sampling policies: sample the edit domain blindly, versus sample a specified non-degenerate fiber.*

(Proof by unfolding the definitions; supplementary Appendix K.2.) The SMS-to-MS degeneration of Theorem 3.1 (Section 3.3.6) is the metric-level shadow of (ii): when the admitted fiber collapses to the syntactic default, the SMS denominator collapses to the MS denominator.

Theorem 3.5 (duality with strong MRs). *Let* r *be a metamorphic relation whose violation set is closed under* $\equiv_\alpha$ *(a* strong *MR). Then* r *detects no semantically inactive mutant, and conversely any mutant detected by a strong MR is semantically active. Hence over a strong MR universe the kill matrix is supported exactly on the semantically active fibers, and inactive (syntactic-only) mutants are structural noise for strong MT evaluation rather than hard cases.*

(Proof by the $\equiv_\alpha$-closure argument; supplementary Appendix K.2.) The duality identifies α as the single hinge connecting MR-side quality (strength) and mutant-side quality (activity). Classification is relative to $(I, \alpha, \text{budget})$: extending I can only move a mutant from active-off-taxonomy to ψ-stratified, never the reverse, so every certificate carries its $(I, \alpha, \text{budget})$ stamp.

Definition (strong MR, weak MR, strong boundary). Because a discrete program P reproduces a continuous operator's invariant only up to a discretisation residual, closure under $\equiv_\alpha$ holds only within a tolerance ε_{tol}. Fixing ε_{tol}, call r *strong* on P when the correct P's residual stays below ε_{tol} (so any violation beyond it certifies a semantically active mutant and Theorem 3.5 holds operationally), and *weak* when the correct P's residual exceeds ε_{tol} (so the correct program is itself flagged by the discretisation, not a fault). The *strong boundary* is the $(\varepsilon_{\text{tol}}, \text{resolution})$ locus separating the two regimes; tightening ε_{tol} or coarsening the discretisation carries a strong MR across it into the weak regime. The full contingency argument is in supplementary Appendix K.2.

Proposition 3.6 (the two boundary failures and their ε_{tol} coupling). *Relative to* $(\varepsilon_{\text{tol}}, P)$ *the duality of Theorem 3.5 has two failure modes. (i)* False positive (weak MR). *When* r *is weak the correct* P *violates* r *and a non-fault is reported; the rotational-symmetry invariant of a neutron-transport operator under a method-of-characteristics discretisation, satisfied only at special angles, is the canonical instance. (ii)* False negative (surviving non-equivalent mutant). *A semantically active mutant* $P_e \not\equiv_\alpha P$ *can leave every invariant in* I *satisfied within* ε_{tol}*, a* coincidental satisfaction of the MR*, and so survive the strong MR family; this surviving mutant is non-equivalent (genuinely* $\not\equiv_\alpha P$*) and is therefore distinct from the classical equivalent mutant of Proposition 3.4(ii), and when its signature lies below* ε_{tol} *it is a* tolerance fault*. The two modes are coupled through* ε_{tol}*: tightening it shrinks the false-negative set (more surviving mutants are killed) but grows the false-positive set (more correct programs are flagged), and loosening it does the reverse. The boundary is therefore an* ε_{tol}*-governed* sensitivity–specificity trade-off*, not a defect to be engineered away.*

Table 4. PUT selection, class summary (12 PUTs across four classes; per-PUT names, mathematical structure, and LOC in supplementary Appendix K.6).

Class	PUTs	Mathematical structures	LOC
A Numeric	A1 Lorenz, A2 LU, A3 FDM	nonlinear ODE, linear algebra, parabolic PDE	80–200
B Probabilistic	B1 Beta-Binomial, B2 MCMC, B3 Monte Carlo	analytic posterior, Markov chain, importance sampling	60–250
C Surrogate	C1 GPR, C2 PCE, C3 NN-surrogate	kernel methods, orthogonal basis, MLP substitution	250–400
D ML	D1 MLP, D2 SVM, D3 Logistic Regression	backprop., convex optimisation, maximum likelihood	120–350

(Proof by threshold-monotonicity in ε_{tol}; supplementary Appendix K.2.)

Operationally, mode (i) does not inflate SMS: the killed predicate requires every counted MR to pass on the unmutated program, so a weak MR is rejected upstream at MR validation and contributes no kills. Proposition 3.6(i) therefore manifests as MR-validation failure on the correct program, and only mode (ii) enters the SMS denominator-numerator arithmetic directly.

Theorem 3.5 thus holds within the strong regime, and the ε_{tol}-governed strong boundary, where a strong MR degrades into a weak MR (false positives) or is evaded by a surviving non-equivalent mutant (false negatives), is mapped empirically on real and textbook programs in Section 5.6.

4 Study Design

The study design follows three validation paradigms rather than relying on a single small experiment. First, analytical validation establishes that SMS is a conservative extension of classical MS. Second, the controlled 12-PUT, 60-cell audit tests constructibility, syntactic separation, aligned-versus-cross discrimination, and root-cause attribution under a reproducible protocol. Third, boundary, adjoint, and real-defect evidence check whether the construct behaves as expected outside the main synthetic-mutant grid. The design is intentionally bounded: compact kernels make semantic certification inspectable, while industrial transfer is treated as a future validity question rather than assumed.

4.1 Program Selection

The 12 PUTs are taken from the shared 12-PUT infrastructure [40] but independently justified along four dimensions: library-stack coverage (numpy 2.4.4, scipy 1.17.1, scikit-learn 1.8.0); mathematical-structure coverage (8 of 12 chapters of *Numerical Recipes*); overlap with existing mutation-testing benchmarks (DeepCrime, Defects4J, and the mutmut and cosmic-ray demos); and signature-simplification trade-offs. The full coverage argument is in Appendix B.1. The `program(x: float) → float` signature is a substantive constraint (Section 8) that bounds the upper limit of mutant semantic complexity; transfer to industrial multi-output PUTs is left to outside the evaluated scope of this paper.

4.2 Necessary conditions for semantic mutation

Definition (Semantic mutation criteria). A mutant `s' = mut_j(S_i)` is a *semantic* mutation if and only if it satisfies at least one of the following three conditions.

(1) **Cross-function-boundary replacement.** The AST node operated on crosses at least one function-call or module-import boundary. For example, `np.linalg.det(M) → np.sum(np.diag(M))`.

(2) **Carries domain knowledge.** The legality of the mutation depends on mathematical, physical, or statistical knowledge of the program's domain, not on purely syntactic type preservation. For example, changing the Gaussian Process Regression `noise_level` from 1e-4 to 1e-1.

(3) **Changes algorithmic class.** The mutation alters the algorithmic class implemented. For example, replacing RK4 with Euler changes the integration order.

A mutation that satisfies none of (a) – (c) is purely *syntactic*: AST-local, domain-agnostic, and class-preserving. The five operator families (CE, OS, HP, TF, SI) are specialisations of (a) – (c):

Table 5. Necessary conditions for semantic mutation.

Operator class	(a)	(b)	(c)	Primary condition
CE Constant perturbation	×	△	×	partial (b); weakest
OS API replacement	✓	✓	△	(a)+(b)
HP Hyperparameter	×	✓	△	(b)+partial(c)
TF Numerical transform	△	✓	✓	(b)+(c)
SI / CF Structural injection	△	✓	✓	(b)+(c)

Only **CE partially satisfies** the necessary conditions (it is a semantic / syntactic boundary class); OS, HP, TF, SI strongly satisfy at least one of (a), (b), (c).

4.3 Metamorphic-Relation Coverage Matrix

Table 6. Metamorphic-relation coverage matrix: per-PUT coverage density across the five meta-patterns (12 PUTs × 5 meta-patterns; MP1–MP5 = $m_{\text{inv}}, m_{\text{mono}}, m_{\text{conv}}, m_{\text{dyn}}, m_{\text{cmp}}$). •• substantial (32 cells), • moderate (19 cells), ∘ vacant (9 cells; no MR is exercised).

PUT	Conservation	Monotonicity	Convergence	Trajectory	Partial order
A1 Lorenz	••	•	••	••	∘
A2 LU	••	∘	•	•	••
A3 FDM	••	•	••	••	∘
B1 BetaBin	••	•	∘	∘	•
B2 MCMC	•	••	••	••	•
B3 MC	••	∘	••	•	∘
C1 GPR	•	••	••	•	••
C2 PCE	••	•	••	•	••
C3 NN-Surr	•	••	••	••	••
D1 MLP	••	••	•	•	••
D2 SVM	•	••	•	∘	••
D3 LR	••	••	•	∘	••

Legend: •• substantial (32 cells); • moderate (19 cells); ∘ vacant (9 cells; no MR exercised).

Each PUT averages 24.3 mutants in the cross-source pool (range 10-30), for 292 unique v4 mutants across the 12 PUTs; each mutant is evaluated in its relevant cells over the 60-cell design with N=20 AVP repetitions per cell, and K_eq

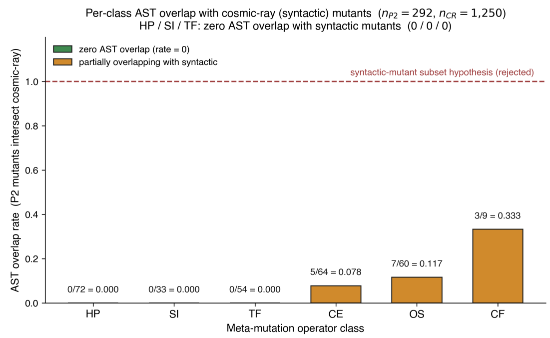

Fig. 2. Per-class AST overlap rate between 292 semantic mutants and 1,250 cosmic-ray syntactic mutants across 12 PUTs (overall 5.14%). HP, SI, TF (54.5% of the semantic pool) show zero overlap (0/72, 0/33, 0/54): SI and TF are structurally cross-function, while HP zero overlap reflects the default first-order value menu; CE 7.81%, OS 11.67%, CF 33.33% (CF is a class-specialisation, not in main 5).

= 1000 input samples for E2 evaluation. Detailed pool counts and per-class operator specialisations are in **Appendix B.2**.

The cell density notation distinguishes ••substantial (32 cells; aligned slice + strong off-diagonal cells where MR coverage is dense and mutant detection is expected), •moderate (19 cells; cross slice with non-trivial expected detection rate), and ∘ vacant (9 cells; historically empty cells where no MR is exercised); the counts are read mechanically from the coverage grids recorded in the MR implementation files (32 substantial, 19 moderate, 9 vacant). The 12 aligned (j = k) cells are defined by each PUT's class-primary meta-pattern; the 48 cross (j ≠ k) cells comprise the remaining substantial, moderate, and vacant cells. The •moderate / ••substantive distinction does not change SMS computation; it tracks expected MR coverage density per the infrastructure documentation and informs the Section 5.9 R_sem / R_kill decoupling discussion.

Per-class operator specialisations. Each operator family (mut_C / M / G / T / F = CE / OS / HP / TF / SI) requires PUT-class-specific specialisation, for example mut_C omits the LU $(k+1)$-th row multiplier or the GPR positive-definite diagonal; mut_G swaps Lorenz RK4 for a 1.5-order hybrid; HP retargets GPR `noise_level` on class c and MLP α on class d; OS swaps `det` ↔ `sum(diag)` on class a. The complete per-family, per-PUT specialisation catalogue is reproduced in supplementary Appendix K.5, and the full PUT-class × operator grid is in Appendix B.2.

4.4 Primary Meta-Pattern Convention

The diagonal cells `j = k` form the H2-aligned slice, the off-diagonal cells form the cross slice, and the vacant cells ∘ are not formally adjudicated.

The c-class primary MP is held at the pre-registered partial-order meta-pattern (m_{cmp}) throughout, and all H1–H4 verdicts are rendered under this configuration on v3 (same-source) and v4 (cross-source, identical prompt). An earlier exploratory data-driven primary-MP shift was withdrawn after a permutation null (10,000 permutations, seed 42, over the 15 exchangeable c-class cell SMS values) showed its δ inflation indistinguishable from random reselection: observed c-class aligned mean 0.314, null mean 0.349, one-sided $p = 0.989$ (`data/results/c_class_permutation_v4.json`); the prohibited selection-on-the-response path (v3b) is thus deferred to a pre-registered rule on a fresh dataset.

4.5 Abstract-Syntax-Tree Overlap against Syntactic Mutants

Experimental design. For each PUT, semantic cross-source mutants (`data/mutants/${PUT}_pool_v4/`, 292 total) are compared at the AST-normalised level (`ast.dump(annotate_fields=False, include_attributes=False)`) with cosmic-ray default-operator mutants (1,250 total; `scripts/p2_vs_syntactic_ast_diff_batch.py`). Source: `data/results/cosmic_ray_12put_ast_diff.json`.

Aggregate results.

Table 7. Semantic vs syntactic mutants, 12-PUT empirical: per-operator-class AST overlap. Aggregate over 12 PUTs: 292 semantic mutants vs 1,250 cosmic-ray syntactic mutants, AST-normalised overlap 15, **overall rate 5.14%**.

Class	n_p2	n_overlap	Rate	Interpretation
HP	72	0	**0.000**	Value-menu artifact (not structural)
SI	33	0	**0.000**	Structurally cross-function
TF	54	0	**0.000**	Structurally cross-function
CE	64	5	0.078	Boundary class (Section 4.2 partial (b))
OS	60	7	0.117	Partial incidental hits
CF	9	3	0.333	b2 only, n=9
All	**292**	**15**	**0.0514**	94.86% AST-disjoint

Interpretation: refuting the "post-classification copy" challenge. HP, SI, and TF (159 of 292 = 54.5% of v4 mutants) show **zero overlap** with cosmic-ray's default AST-local operators: for SI and TF the gap is structural (cross-function edits no first-order operator expresses), while HP's zero overlap is a value-menu artifact (a single-literal edit whose replacement value the default menu does not enumerate), not a structural impossibility, and whether HOM compositions [33] close it is a residual threat (Section 4.6). CE shows 7.81% incidental overlap (mostly half-step perturbations such as `_RHO=28.0` → `27.5`), the remaining 92.19% AST-disjoint; OS refines an earlier "tool-inexpressible" label to 88.33% disjoint / 11.67% incidental hits (stochastic byproducts, not systematic mutation; Section 4.6, Appendix B.1.5). Overall 94.86% of v4 mutants cannot be reproduced by cosmic-ray defaults, so the two pools occupy systematically distinct mutant spaces.

A multi-tool cross-comparison was not run because mutpy is incompatible with Python 3.10+, and mutmut's operator set overlaps strongly with cosmic-ray's. The DeepSeek, Claude, and GPT contributions to the 15 overlap files are uneven (DeepSeek 11, Claude 4, GPT 0). Appendix B.3 covers the cosmic-ray a1 single-PUT pre-12-PUT pilot, and Appendix F discusses the LLM-source distributional-shift threat.

4.6 Higher-Order Mutation Boundary

Scope of the claim. The preventive-defence claim below is conditional on a first-order syntactic baseline; HOM-based syntactic compositions are an open question and are not refuted by the Section 4.5 evidence.

The HP, SI, and TF zero-overlap result, combined with the Section 4.2 necessary-conditions argument, amounts to a *preventive-defence* argument: semantic mutation operators address a class of fault hypotheses that lie beyond the reach of default first-order syntactic configurations (HOM not refuted). Three points sharpen this framing.

(a) Systematic versus incidental. Satisfying conditions (a), (b), or (c) is sufficient for a single semantic mutation, but only when satisfaction reflects design intent, not stochastic byproduct, does it constitute a *systematic* semantic mutation method. A syntactic tool that occasionally hits (a) or (c) with its 13 default operators does so at a non-zero probability, but the hits are not repeatable and they carry neither of the two engineering goods we want. First, designing a semantic mutator like OS `det` → `sum(diag)` requires knowing that these expressions are equivalent on diagonal matrices but not on general matrices, a deepening of source-code understanding the syntactic tool does not exhibit. Second, domain-semantic faults, wrong physical constants, unit conversions, boundary conditions, hyperparameter semantics, numerical-method order, are not AST-local; the syntactic-tool design goals (operator typos, off-by-one errors, negation flips) do not target them. Systematic semantic mutation therefore requires (a), (b), or (c) to be design intent. Appendix B.1.5 records this argument in full.

(b) HOM caveat. The Section 5 empirics did not run a HOM comparison, so we confine the "tool unreachability" claim to *first-order* syntactic tools (mutmut and cosmic-ray default configurations) and list HOM equivalence testing as a residual threat (Appendix F.2). Whether HOM [34, 38] compositions can partially simulate OS or HP effects is analysed conceptually next.

Conceptual analysis of HOM reachability (summary). Higher-order mutation (HOM; 33, 34, 38) composes multiple first-order operators on one source location. Analysed condition by condition (full argument in supplementary Appendix K.3), default-operator HOM cannot systematically reach any of the three necessary conditions: (a) cross-boundary swaps like `det(M)` → `sum(np.diag(M))` presuppose a domain-specific equivalence no AST-local chain introduces; (b) HOM is unaware of which constants are load-bearing, whereas HP is parameter-aware by construction; and (c) algorithmic-class change requires structural rewrites beyond AST-local edits. We therefore assert that the Section 4.5 0/0/0 unreachability for HP / SI / TF, while strictly empirical only for first-order configurations, is unlikely to be refuted by default-operator HOM compositions on combinatorial grounds; a direct empirical HOM falsification remains a residual threat. A multi-tool cross-comparison (mutmut, mutpy) was not run (mutpy is incompatible with Python 3.10+, mutmut's operator set overlaps cosmic-ray's); the aggregate of the two default sets (~21 first-order AST-local operator classes) collectively fails all three necessary conditions (cross-reference in Appendix B.6). The 15 overlaps are also LLM-source-skewed (DeepSeek 11/15, Claude 4/15, GPT 0/15; distributional-shift threat in Appendix F.2), but this does not affect the categorical argument: the HP / SI / TF result is 0/0/0 across all three sources.

4.7 Adjoint Extension Subjects

The adjoint invariant ψ_6 (Section 3.3.7) is exercised on a separate, confirmatory set of three third-party linear-operator libraries: scipy `LinearOperator` (linear algebra), pylops real-FFT (signal processing), and jax `linear_transpose` (automatic differentiation), disjoint from the 12 PUTs and the pre-registered 60-cell. Each library carries a real fixed defect in its adjoint code path as a ground-truth anchor (scipy #8900, pylops #292, jax #6223). For each, the MR families are adj.self (self-adjointness) and adj.dual (scaled/dual-operator transpose), and the mutant pool mixes adjoint-breaking mutants (the real defect among them) with semantically equivalent mutants a strong MR must *not* kill; the real defects are reproduced in extracted-diff form because the pre-fix releases are uninstallable on the host (no `arm64` wheel). The arm is confirmatory and additive: it does not alter the 60-cell denominator, the primary-MP convention, or the pre-registered hypotheses. Full subject specifications and fault mechanisms are in supplementary Appendix H; results in Section 5.7.

4.8 Empirical Study Procedure

With the subjects, operators, and the adjoint extension arm fixed (Sections 4.1–4.7), we now specify the procedure that produces each cell's measurements.

Pipeline order: Section 4.1 PUTs → Section 4.9 Mutant generation → LRCA L0 prescreen → Section 4.11 Equivalence ($E_1 \wedge E_2$) → AVP → killed/survive → LRCA three-layer diagnosis → SMS + C_1_share report.

Each (i, k, j) cell is reported with `inst_count`, `equiv_count`, `killed_count`, `survive_count`, SMS_{i,k,j}, C1_share_{i,k,j}, suspect_share_{i,k,j}, and the three rates `inst_rate / equiv_rate / survive_rate` corresponding to RQ3.

4.9 Cross-Source Generation Protocol

To isolate the contributions of LLM same-source bias and MR-MP alignment design to Cliff's delta, we ran a three-stage ablation:

Table 8. Cross-source protocol.

Version	Mutant pool	c-class primary MP	Use
v3	Same-source (Claude Opus 4.6)	partial-order (pre-registered)	H2 baseline (pre-registered primary)
v4	**Cross-source** (Claude Opus 4.6 + GPT-5.4 + DeepSeek chat)	partial-order (held at pre-registered)	Isolate LLM-source diversity

The v4 protocol (`scripts/cross_source_campaign.py`) runs each (PUT, operator) pair on Claude / GPT / DeepSeek with K=3 trials under an identical prompt template (temperature 0.7, V1-V4 mechanical-validation gate). Cross-source pool capacity: 37 operators × 3 sources × 3 trials = 333 attempts, 89.5% V1-V4 pass, 298 confirmed mutants distributed nearly equally (Claude 101, GPT 98, DeepSeek 99); final-stage rejection of duplicate and QA-failing mutants leaves the 292 that enter the v4 pool used for SMS and the AST audit. Here "37 operators" denotes the executable operator-specialisation inventory: 36 primary PUT-class operator pairs plus one CF structural-injection specialisation. The conceptual operator family count remains five (CE, OS, HP, TF, SI); CF is reported only as a specialised SI/structural-injection row in the AST-overlap audit.

Declared confound: protocol asymmetry. v3 used the original Phase-1 dual-blind reviewer protocol (Claude generation + GPT-5.4 review + DeepSeek arbitration); v4 passes V1-V4 mechanical gates only and does **not** invoke a reviewer LLM. A fraction of the v3 → v4 source-axis contrast may therefore reflect a slight quality shift rather than LLM-source diversity per se. The follow-up study will rerun dual-blind on the v4 grid, but at $n \geq 30$ PUTs: a same-n rerun removes the protocol-asymmetry confound yet, at $n = 12$ PUTs, the $\Delta\delta$ CI remains ≈ 0.44 wide, so detecting a $\Delta\delta = 0.20$ source-diversity effect at 80% power requires ≈ 30 PUTs, and the follow-up is therefore designed at that size. Full per-LLM token / latency / cost figures, V1-V4 specifications, and the full differential-prompt protocol are in **Appendix C.1**.

4.10 Mutant-pool prescreen and equivalence detection

Pool prescreen (LRCA L0). Each candidate passes three gates, static lint/type checking, a unit self-test for finite output, and a Phase-1 double-blind review sign-off (Claude Opus generator, isolated GPT-5.4 reviewer emitting a syntactic/executable/fault-injected triple; inconsistent outputs go to a $\leq 10\%$ manual arbitration queue), with a further 20% manual re-review that downgrades the batch if manual–reviewer inconsistency exceeds 10%. Each cell produces

30–50 candidates and retains 10–15. The v4 cross-source pool follows a different L0 path (V1–V4 mechanical validation, duplicate removal, QA rejection) that does **not** invoke the reviewer LLM for cost reasons, so v4 C5 (artefact) labels are assigned only at LRCA artefact recheck by a human reviewer, not inherited from a blind vote; this protocol asymmetry (a possible v3→v4 quality decline) is declared in Section 5.8, with full justification in Appendices C.1–C.3.

Equivalence detection (E1 ∧ E2). For each mutant $s' \in \text{mut}_j(S_i)$:

(1) **E2 step.** Sample $K_{\text{eq}} = 1000$ inputs $x \sim D_S$; compute $\|S_i(x) - s'(x)\|$; if $\leq \varepsilon_{\text{eq}}$ for all samples, mark E2-pass.
(2) **E1 step.** For every $mr \in \text{MR}_{i,k}$, compare $\text{AVP}(S_i, mr)$ with $\text{AVP}(s', mr)$; if all consistent, mark E1-pass.
(3) **Conjunction.** E1 ∧ E2 → enter $\text{equiv}_{i,k,j}$, exclude from SMS denominator. Otherwise pass to AVP-killed determination.

E2 is run before E1 because E2 is computationally cheaper (K_eq scalar evaluations) and quickly discards numerically distinct mutants; the remaining candidates are screened by AVP for MR-relation equivalence. The joint condition is conservative: false-non-equivalent (E1 ∨ E2 fail) is much easier than false-equivalent (both must mis-pass simultaneously), biasing SMS slightly *low*: truly equivalent mutants misclassified as non-equivalent remain in the admitted denominator as unkillable survivors, understating adequacy, an explicit conservative engineering choice (Appendix A.3).

Killed determination. With OR-aggregation across mr in MR_{i,k}:

$$\text{killed}(s', \text{MR}_{i,k}) \iff \exists\, mr \in \text{MR}_{i,k} : \\ \text{AVP}(S_i, mr) = \text{pass} \wedge \text{AVP}(s', mr) = \text{fail}.$$

Each (i, k, j) cell is sampled N = 20 times (statistical replicate) to compute the per-cell SMS, with mutant-level fail ratios feeding the LRCA L1 decision. The AVP version is pinned by the frozen dependency file and the embedded source in the replication package; the reproducibility package carries the complete AVP source so that the package remains self-consistent under any upstream evolution.

4.11 Three-Layer Root-Cause Diagnosis

For each killed mutant the three-layer decision tree of Section 3.3.4 applies in order (L1 tolerance robustness, all classes; L2 OOD triage, C/D; L3 statistical-assumption baseline, B/D; then artefact recheck), under priority C5 > C4 > C3 > C2 > C1. Threshold calibration over a 9-grid (`ood_band` $\in \{0.02, 0.05, 0.10\} \times$ `tolerance_multiplier` $\in \{3.0, 10.0, 30.0\}$, repeats 20) lifts H4 from 10/60 to 12/60 cells; the best combination (ood_band 0.02, tolerance_multiplier 3.0) is primary, with default-threshold results as control (`lrca_60cell_v3.json`). The 12/60 = 20% ceiling stays far below the 80% threshold, so H4 is unattainable on this dataset as an intrinsic property of LLM-mutant pools. Detailed grid, L0–L3 sub-protocols, and pseudocode are in **Appendix A.2 + C.4**.

5 Results and Discussion

The empirical thresholds H1–H4 are stress tests of this particular semantic-mutant instantiation, not prerequisites for the definition of SMS. The positive validation target is narrower and more structural: SMS must preserve the mutation-score denominator, separate domain-semantic mutants from default first-order syntactic mutants, make MR adequacy failures diagnosable, and distinguish mutant kill-rate, semantic-stratum alignment, and real-defect detection. Thus the failed thresholds below are reported as boundary findings about operator applicability, MR design, and kill attribution; they do not invalidate the metric construction.

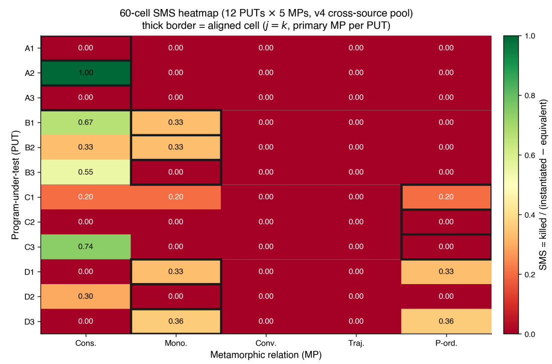

Fig. 3. 60-cell heatmap of mean SMS over 12 PUTs × 5 MPs (v4 cross-source primary). Diagonal cells (operator-MP aligned) show concentrated mass; off-diagonal mass reflects bleed into adjacent MPs. Empty rows = PUT-level zero-mass cohort (PUTs A1, A3, C2; see Section 5.9).

5.1 Statistical pipeline

The primary statistical reporting follows a pre-registered hierarchy:

Table 9. Statistical pipeline.

RQ	Primary statistics	Reporting format
RQ3	inst_rate, equiv_rate, C1_share, survive_rate	60-cell heatmap + 4-class marginals
RQ4	aligned-SMS vs cross-SMS; sparse ∘ vs dense ••equiv_rate	Cliff's delta + odds ratio + 95% bootstrap CI
RQ5	ΔSMS_c (c ∈ {A,B,C,D}); CV(ΔSMS)	directional consistency count over the 4 classes (no p-value attached) + descriptive forest plot
RQ5 (desc.)	Spearman ρ + Kendall τ (SMS vs PC)	scatter + dual-metric ranking comparison

This pipeline answers the empirical questions RQ3–RQ5; RQ1 is answered by Theorem 3.1 (Section 3.3.6) and RQ2 by the AST-normalised structural audit (Section 4.5), outside the statistical pipeline.

Where multiple per-class tests are reported (the per-class Friedman tests of Section 5.11), they carry a Bonferroni correction across the four classes; we do not report a separate cell-level family of p-values, so no family-wise cell-level correction is claimed. Bootstrap 95% CIs use 1,000 iterations as default (B = 10,000 for the headline H2 delta CI). Mixed-effects modelling (`sms ~ C(class) * C(operator) + (1 | put)`) failed with a Singular matrix at N = 60: the class × operator interaction design matrix is rank-deficient and the PUT random-intercept variance collapses to the boundary at this sample size. The Friedman test is the non-parametric formal alternative. The four hypotheses are pre-registered: H1 (operator implementability) requires ≥ 4 of 5 operators producing ≥ 5 non-equivalent mutants on $\geq 9/12$ PUTs; H2 (aligned-vs-cross) requires odds ratio ≥ 3.0 *and* Cliff's delta ≥ 0.474 (Romano 2006 large-effect threshold); H3 (cross-class consistency) requires sign-test 4/4 across 4 classes plus CV(ΔSMS) < 0.5; H4 (LRCA mass) requires mean suspect_share ≤ 0.20. SMS variants (SMS, SMS_unfiltered) and construction lemmas are in **Appendix D.1**.

The results are reported in RQ order. RQ3 is answered in Section 5.2. RQ4 spans the aligned-versus-cross comparison (Section 5.3), its power analysis (Section 5.4), the root-cause diagnostics (Section 5.5), the strong-boundary and adjoint studies (Sections 5.6 and 5.7), the cross-source analysis (Sections 5.8 and 5.9), and the industrial construct-separation arm (Section 5.10). RQ5 follows in Sections 5.11 and 5.12, with the class-level stability discussion in Section 5.13.

Because the 60 cells share 12 PUTs and are not independent, each reported statistic carries an explicit inference permission; no number in this paper should be read above its row. Table 10 is the single cross-study verdict-and-permission ledger: it lists every hypothesis across the three studies with its pre-registration status, sampling unit, verdict, and licensed reading, and *subsumes* the per-study scoreboards (the Study-2 and Study-3 key statistics are reported in the prose of Sections 6.4 and 7.3).

Table 10. Cross-study inference-permission and verdict ledger (all 11 hypotheses plus the descriptive statistics), spanning Study 1, Study 2, and Study 3. This table subsumes the per-study scoreboards.

Statistic	Pre-registration status	Sampling unit (n)	Verdict	Licensed reading
H1 operator–PUT counts (S1)	pre-registered threshold	PUT × operator	NOT MET	deterministic count; verdict factual on this pool
H2 Cliff's δ, odds ratio, bootstrap CI (S1)	pre-registered exploratory large-effect target	cell (12 aligned vs 48 cross; non-independent within PUT)	NOT MET	point-estimate verdict only; no population claim (pass probability ≈ 0.5 at the boundary is rule-intrinsic; Section 5.4)
H3 sign test, CV(ΔSMS) (S1)	pre-registered threshold	class (n = 4)	NOT MET	verdict factual on these four classes
H4 mean suspect_share (S1)	pre-registered threshold	cell (n = 60)	NOT MET	descriptive mean; no independence assumed
Friedman χ^2 (MP effect; per-class Bonferroni) (S1)	exploratory inferential	PUT blocks (n = 12)	sig. (expl.)	non-parametric alternative after mixed-effects singularity; exploratory
Spearman ρ / Kendall τ (SMS vs PC) (S1)	none (descriptive)	PUT (n = 12)	descriptive	hypothesis-generating only
Industrial arm: four-group Wilcoxon, Holm-corrected	pre-registered in the dataset protocol	case (n = 34, paired)	T1>B1 holds	inferential within the 34-case pool; no corpus-level claim
Industrial arm: 34/34 real-defect face	none (descriptive; selection-conditioned)	case	34/34	face-level count; not a coverage rate
H2-1′ Cliff's δ (S2)	pre-registered confirmatory (Family A)	cell (28 aligned vs 112 cross; non-independent within PUT)	CONFIRM	one-sided lower bound > 0 licenses directional dominance; Romano band descriptive only
H1′ family counts (S2)	pre-registered confirmatory (Family E)	PUT × family	CONFIRM	deterministic count; verdict factual on this pool
H3′ class direction (S2)	pre-registered confirmatory (Family E)	class (n = 4)	CONFIRM	direction count; verdict factual on these four classes
H4′ mean suspect_share (S2)	pre-registered confirmatory (Family E)	cell (n = 140)	NOT_CONF.	descriptive mean; verdict factual (threshold not met)
H2-2 cross-vendor $\Delta\delta$ (S2)	registered, gated (Family B)	n/a (not run)	NOT-RUN	not-run under same-vendor freeze; cross-vendor question open, not a null
H4″-graded rich-class mean share (S3)	pre-registered confirmatory (Family G, Holm 2)	detected mutant, aggregated over rich C/D PUTs ($n_{\text{rich}} = 6$)	NOT_CONF.	one-sided lower bound ≤ 0.15; verdict factual (NOT_CONFIRMED); no purity claim on rich classes
H4″-strict single-stratum purity (S3)	pre-registered confirmatory (Family G, Holm 2)	detected clean-family mutant ($n = 90$)	CONFIRM	one-sided lower CP bound ≥ 0.90 licenses single-stratum validity on {CE, HP, CF-with-screen} only

5.2 RQ3: Semantic-Mutation Score Distribution

Table 11. RQ3: Semantic-mutation score distribution.

Metric	Value
Number of cells	60
Mean SMS	0.104
Median SMS	0.000
Std SMS	0.213
Cells with SMS = 0	45 / 60
Mean mutants / PUT (v4)	24.3 (range 10-30)

The **zero-mass dominance** (45/60 = 75%) is concentrated in the cross-MP slice: ~88% (42/48) of cross cells are zero, ~25% (3/12) of aligned cells are zero. Cliff's delta is well-defined under this distribution but its inference is effectively dominated by `n_aligned_nonzero ≈ 6 + n_cross_nonzero ≈ 9 = 15`, not the surface n=60. Three odds-ratio-like quantities must be kept distinct. First, the pre-registered *median* odds ratio is formally $+\infty$ because the cross-slice median SMS is exactly 0; it degenerates under zero-inflation and licenses no verdict, so we report "aligned median > 0 = cross median" only as auxiliary qualitative evidence. Second, a post-hoc *detection-incidence* sensitivity, declared in its own single-test family (outside the pre-registered Holm family and outside the H2 pre-registration), binarizes SMS and asks how often each slice scores any nonzero effect. Under the MP5-held primary convention, aligned cells are nonzero in 6/12 cases versus cross cells in 9/48 (incidence 50% vs 18.75%; only 9 of the 12 PUTs carry any nonzero cell). A one-sided Fisher exact test gives $p = 0.035$ with conditional-MLE odds ratio 4.2 (sample OR 4.33; exact one-sided 95% CI $[1.12, +\infty)$), and the advantage is directionally stable across pool variants (OR 4.1–7.0, one-sided $p = 0.006$–0.049). Aligned meta-patterns therefore detect a nonzero effect appreciably more often than cross patterns, a distinct and modest incidence separation, not the near-twentyfold ratio that a PUT-level any-signal count (9 of 12 PUTs non-dead) would suggest if mislabelled as an aligned-cell count. Third, the Cliff's δ *magnitude* estimand (Section 5.3) is the pre-registered H2 criterion: incidence separates, magnitude does not, and the H2 verdict remains **not met**.

Effective-n note. The surface $n_{\text{aligned}} = 12$, $n_{\text{cross}} = 48$ mask a much smaller information content: only about 15 observations carry nonzero SMS (6 aligned + 9 cross). This heuristic count (not a formal effective sample size) explains the wide 95% bootstrap CI [0.014, 0.622] of the v4 primary delta, consistent with the liberal tendency of the percentile bootstrap when few observations carry signal. Power and effect-size disambiguation are treated jointly with the stipulated-alternative analysis in Section 5.4, and PUT-class diversification at $n \geq 30$ is a testable route to relaxing the constraint.

H1 verdict: not met. H1 (RQ3) requires at least 4 of the 5 operators to produce ≥ 5 non-equivalent (confirmed) mutants on at least 9 of the 12 PUTs. Table 12 counts confirmed non-equivalent mutants per operator class across the 12 PUTs: only the Hyperparameter operator clears the $\geq 9/12$ bar.

Table 12. H1: PUTs reaching ≥ 5 non-equivalent mutants, per operator class.

Operator class	PUTs with ≥ 5 non-equiv mutants (of 12)
Conservation Erosion (CE)	4 / 12
Operator Substitution (OS)	5 / 12

Hyperparameter (HP)	**9 / 12**
Trajectory Flip (TF)	5 / 12
Structural Injection (SI)	1 / 12

Because the semantic operators are class-targeted, each is applicable to a subset of the four classes, not uniformly to all twelve; only Hyperparameter clears the 9-PUT threshold, so 1 of 5 meets the criterion and **H1 is not met as pre-registered**, the uniform-applicability assumption behind the $\geq 9/12$ bar failing for class-targeted operators. A secondary applicability-adjusted denominator (counting only PUTs with a class-specific specialisation, Appendix B.2/C.4) gives CE 4/8, OS 5/7, HP 9/9, TF 5/6, SI 1/6, which does not overturn the verdict but locates its source: HP and TF are implementable across their intended classes, OS is moderately broad, and SI remains a narrow high-risk family (per-operator profile in Appendix C.4).

5.3 RQ4: Aligned and Cross-Pattern Comparisons

Table 13. RQ4: Aligned and cross-pattern comparisons.

Slice	n	Mean SMS	Median SMS
aligned (j = k)	12	0.213	0.100
cross (j ≠ k)	48	0.077	0.000

The means and medians above are those of the MP5-held cross-source pool (`rq2_cliffs_delta_v4_mp5.json`) that generates the headline $\delta = 0.314$ reported below, so the summary statistics and the effect size share one provenance. (The unconditioned v4 pool with the c-class primary at MP1 gives higher means 0.275 / 0.061 and $\delta = 0.439$; the power analysis in Section 5.4 is run on that unconditioned pool, as noted there.)

Two delta point-estimates under the pre-registered primary-MP convention:

- **v3 (same-source, pre-registered):** delta = **0.323**, 95% CI [0.017, 0.622]
- **v4 (cross-source, c-class held at partial-order):** delta = **0.314**, 95% CI [0.014, 0.622]. The cross-source pool aggregates Claude / GPT / DeepSeek under an identical prompt; the c-class primary is held at the pre-registered partial-order meta-pattern so that the v3 → v4 contrast holds the prompt and primary-MP choice fixed. Because v4 uses mechanical gates rather than the v3 dual-blind reviewer step, the source-axis reading is qualified by the protocol-asymmetry analysis in Section 8.3.

H2 verdict: not met. H2 carries two pre-registered criteria, Cliff's $\delta \geq 0.474$ and an aligned-to-cross odds ratio ≥ 3.0. Neither delta crosses the Romano et al. [45] large-effect threshold 0.474, 0.323 and 0.314 both fall just below the Romano et al. [45] medium threshold of 0.330, so the delta criterion fails. The Romano threshold is retained as the frozen decision anchor because it is the threshold recorded in the experiment design; we do not re-anchor the verdict post hoc to a Vargha–Delaney interpretation scale. The odds ratio is degenerate: because the cross-slice median SMS is exactly 0 (table above), the median odds ratio is $+\infty$, so the ratio criterion is satisfied only trivially under zero-inflation and carries no evidential weight. H2 is therefore not met as a large-effect target. The contrast $\delta_{v4} - \delta_{v3} = -0.009$ (paired-role bootstrap 95% CI [-0.238, 0.207]) is too small to interpret as a stable LLM-source-diversity effect under the present asymmetric protocol: replacing a same-source pool with a three-LLM cross-source pool does not appreciably shift Cliff's delta within this design, but the exact source-diversity contribution requires a dual-blind v4 rerun. We frame H2 as an underpowered exploratory contribution and defer full verification to a follow-up study with a larger PUT sample.

Vacant-cell sensitivity. Excluding the 9 cells marked vacant in Table 6 gives the same H2 verdict. Under the frozen primary-MR convention, v4 has n = 11 aligned and n = 40 cross cells, $\delta = 0.323$ with 95% CI [-0.011, 0.648]; v3 has $\delta = 0.270$ with 95% CI [-0.043, 0.595]. The direction remains positive but still below 0.474, and the interval crosses zero once vacant cells are removed. The sensitivity is a denominator check, not a changed conclusion.

This medium-effect placement is consistent with Tip et al. [47] LLMorpheus's medium-effect range on JavaScript LLM mutants, a contextual literature observation (estimand caveat: their delta compares "LLM vs traditional mutants on fault detection", ours compares "aligned vs cross MP slice within one pool"; the numerical similarity is not substantive support).

5.4 RQ4: Stipulated-Alternative Power Analysis

The Section 5.2 effective-n constraint motivates an explicit power analysis (full exceedance and stipulated-alternative tables in supplementary Appendix K.4; sample-size sweep in Appendix D.3). A plug-in bootstrap (5,000 replications, seed 42) over the unconditioned ($n = 12$, $n = 48$) pool ($\delta = 0.439$) exceeds the "any effect" threshold with probability 0.997, the medium threshold (0.330) with 0.759, and the large H2 threshold (0.474) with only 0.423. A stipulated-alternative simulation, in which the truth is fixed at the H2 boundary 0.474 (a mixture calibrated to $E[\delta] = 0.4746$, since SMS ties at zero make a raw shift discontinuous), returns the point-estimate pass rule with power **0.491** and the "lower 95% CI bound > 0" rule with power 0.868.

Even when the truth equals the H2 boundary, the point-estimate decision rule returns "not met" in roughly half of replications; a pass probability near one half at the boundary is intrinsic to a threshold rule applied to an approximately median-unbiased estimator at any sample size, so the 49.1% figure characterises the pre-registered decision rule, not a deficiency specific to the (12, 48) design. This supports the framing in Section 5.3: the H2 verdict is a factual statement about the point estimate failing to clear the threshold, not a claim that the effect is necessarily smaller than 0.474. Increasing the sample size narrows the confidence interval but cannot lift the point estimate.

The source-axis contrast is underpowered too. We did not run a separate power simulation for the source-axis contrast $\Delta\delta = -0.009$ (paired-role bootstrap 95% CI [-0.238, 0.207]); its estimand and dependence structure differ from the aligned-vs-cross δ, so the 49.1% figure above does not transfer to it. The small design (n = 12 PUTs) leaves the -0.009 null consistent with a wide range of true source-diversity effects, so we explicitly do not read it as evidence that source diversity is inert. A properly-powered strong-sense test is deferred to a follow-up study.

Note on monotone-transformation invariance. Cliff's δ is rank-based: a function of $U/(n_1 \cdot n_2)$, so applying a logit transform to SMS gives $\delta_{\text{logit}} \equiv \delta_{\text{raw}}$ by construction. This is consistent with the rank-invariance theorem and does **not** constitute additional robustness evidence. The genuine robustness threats to the H2 verdict come from the zero-mass dominance (Section 5.2), not from any metric-scale choice.

5.5 RQ4: Root-Cause Diagnostic Results

H4 verdict: not met. The pre-registered criterion is mean suspect_share ≤ 0.20; the observed mean is **0.791** (v4), decisively above the threshold, so H4 fails on its own metric. A dense cutoff sweep (Appendix D.2) confirms this is an intrinsic property of LLM-mutant pools rather than a calibration artefact: the secondary per-cell pass-ratio (cells with

Table 14. RQ4: Root-cause diagnostic mass.

Metric	Value
Mean C1_share (v3 same-source pool)	0.164
Mean C1_share (v4 cross-source pool)	**0.209**
Mean suspect_share (v4 cross-source pool)	**0.791**
Cells with suspect_share ≤ 0.20 (secondary; threshold-calibration best, ood_band = 0.02)	12 / 60 = 20.0%

suspect_share ≤ 0.20) is flat at 12 / 60 = 20% across cutoffs $\in [0.05, 0.40]$, because v4's suspect_share distribution is severely bimodal (median 1.0, with 48 cross cells near 1 and 12 aligned cells near 0; the [0.20, 0.80] interior is nearly empty). H4 was pre-registered before pool characteristics were known, so this is a finding worth reporting.

C1-weighted sensitivity. Because LRCA deliberately does not filter SMS, we add a diagnostic secondary view $\mathrm{SMS}_{C1} = \mathrm{SMS} \times \mathrm{C1\_share}$. On the v4 60-cell table, mean SMS is 0.104 and mean SMS_{C1} is 0.0873; nonzero cells fall from 15/60 to 13/60. The conclusion is therefore conservative: suspect mass is high and H4 is not met, but the main positive SMS signal is not created solely by suspect labels. We keep raw SMS as the primary metric because LRCA is a diagnostic attribution layer, not part of the mutation-score denominator.

5.6 RQ4: Strong-Boundary Failure Modes

The 60-cell audit (Sections 5.2–5.5) exercises strong MRs in the regime where Theorem 3.5 holds. To map the strong boundary of Proposition 3.6 we run a complementary study on six third-party or textbook programs spanning the meta-patterns, kept separate from the 12 PUTs. Wherever a genuine fixed defect exists in a third-party library we use real version-switching: the pre-fix and post-fix releases run unmodified in isolated environments, and we record the repository, the version pair, and the issue/PR, and we fall back to a textbook faithful implementation only when the real defect cannot be exercised on the host (no `arm64` wheel, a heavyweight Fortran build, or a drift below machine resolution). $\varepsilon_{\mathrm{tol}}$ is an explicit factor in every case.

Table 15. RQ4: Strong-boundary map over six programs.

Family	Program (provenance; method)	Strong MR ($\varepsilon_{\mathrm{tol}}$)	Correct / faulty verdict	Boundary role
inv.con	non-conservative Burgers; [29] (textbook)	shock speed (0.03)	0.498 pass / −0.002 killed	strong
inv.eqv	scipy `align_vectors` 1.13.0 → 1.13.1, #20660 (real)	rotation residual (10^{-6})	10^{-16} pass / 2.0 killed	strong
adj.self	scipy `_adjoint` #8900/PR#8962 (real, extracted-diff)	adjoint-identity residual	10^{-16} pass / 1.4–2.0 killed	strong
rev.traj	leapfrog vs explicit Euler; [28] (textbook)	energy-drift ratio (≤ 3)	1.00 pass / 1952 killed	strong

PINN	soft vs hard BC; DeepXDE #26 (real torch)	exact BC $\lvert u(0)\rvert$ (10^{-4})	hard 0 pass / soft 5×10^{-4} killed	FP (weak MR)
struct. (FN)	numpy float32 RNG 1.21.6 → 1.22.0, #17478 (real)	mean / range / dist	both pass; buggy survives	FN (surviving)

Strong regime (four cases). For inv.con, inv.eqv, adj.self, and rev.traj the correct program satisfies the strong MR within ε_{tol} while the faulty program is killed, exactly as Theorem 3.5 predicts. Two are real fixed library defects (scipy `align_vectors`; scipy `_ScaledLinearOperator._adjoint`, exercised by reverse-applying the fix diff because the pre-fix release has no `arm64` wheel); two are textbook contrasts for defects unexerciseable on the host (Burgers conservative/non-conservative for the GeoClaw conservation bug; symplectic leapfrog vs explicit Euler for the REBOUND IAS15 drift, real signature near 10^{-14} below resolution).

False positive, weak MR (PINN). A soft-penalty PINN is a legitimately trained, non-faulty model, yet its exact-boundary residual $|u(0)| \approx 5 \times 10^{-4}$ exceeds a strict $\varepsilon_{\text{tol}} = 10^{-4}$, so the strong MR flags it: the discretisation (soft constraint), not a fault, breaks the invariant, the empirical face of Proposition 3.6(i). The hard-constraint ansatz $u = x(1-x)\,\mathrm{NN}(x)$ satisfies the invariant by construction and passes.

False negative, surviving non-equivalent mutant (RNG). The numpy float32 defect (#17478) zeroes the lowest mantissa bit; the structural MR family (mean ≈ 0.5, range $[0, 1)$, distribution shape) is satisfied by both releases within ε_{tol}, so the genuinely non-equivalent buggy version survives (coincidental MR satisfaction; an odd-bit-fraction discriminator reads 0.0 for the buggy vs ≈ 0.5 for the correct release, so it is non-equivalent, not a classical equivalent mutant). It is the empirical face of Proposition 3.6(ii).

The ε_{tol} trade-off. The PINN case makes the sensitivity–specificity coupling runnable: at strict $\varepsilon_{\text{tol}} = 10^{-4}$ the soft-BC model is killed (false positive), at loose 10^{-3} it survives, and had the residual arisen from a true fault, loosening would instead cause a false negative. The strong boundary is thus a tunable operating point, the honest scope of the duality, not a defect to remove.

5.7 RQ4: Adjoint Extension Evidence

Where Section 5.6 maps the boundary at which the duality fails, a small confirmatory arm checks it holding in the forward direction for the sixth meta-pattern ψ_6. On the three third-party linear-operator libraries of Section 4.7 (scipy, pylops, jax), each carrying a real fixed adjoint defect, a strong adjoint MR detects only the active mutants: across all three substrates the correct program survives, every non-equivalent mutant (including the three real defects scipy #8900, pylops #292, and jax #6223) is killed, and no semantically equivalent mutant is killed, giving a forward SMS of 1.00 with zero false positives on the equivalent set (full subjects, mutant pools, and the per-substrate kill table in supplementary Appendix H). The arm is confirmatory and additive, it does not enter the 60-cell denominator and leaves H1–H4 unchanged, and carries two scope caveats: the scipy #8900 defect is shared with the Section 5.6 boundary study (so it is not counted as independent corroboration twice), and the pools are hand-constructed, low-cardinality sets with the real defects reproduced in extracted-diff form, so the arm demonstrates rather than measures adequacy (adjoint-arm reproduction-fidelity threat).

5.8 Cross-source contributes mutant quality, not effect size

Going from same-source to cross-source under the pre-registered primary-MP convention shifts Cliff's δ by only -0.009 (95% CI [-0.238, 0.207]) while raising mean C1_share from 0.164 to 0.209 (+27%), class-c mean SMS by **+91.5%**, and class-d by 38.3%. Within this fixed-prompt design, LLM source identity is therefore not the dominant factor behind the H2 ceiling: cross-source pooling improves the diagnostic quality of the kill set without moving the aligned-vs-cross effect size. This is a scoped null, not a claim that LLM diversity is inert; the strong-sense test with per-LLM differential prompts (V_persona, V_cot) is deferred (Appendix C.1). The Section 4.5 evidence (94.86% of v4 mutants AST-disjoint from cosmic-ray defaults; HP/SI/TF at 0/0/0) confirms the medium-effect ceiling is not an artefact of LLM-pool overlap with the syntactic space, so a larger aligned-versus-cross effect points first to MR-design refinement, not more sampling. (As a contextual coincidence, Petrovi'c et al.'s [2021] ~20% productive-mutant rate at Google is close to our calibration-sweep C1_share of 0.20, distinct from the pool values 0.209 (v4) and 0.164 (v3); the constructs differ (developer survey vs kill-behaviour annotation), so this is not mechanism validation.)

5.9 Decoupling between Semantic Feasibility and Kill Rate

The operator-level pilot in Appendix C.4.1 reveals a sharp pattern: HP, TF, and OS operators on the c-class (surrogate) and d-class (ML) PUTs reach R_sem $\approx$ 1.0 (a valid, executable, intent-bearing mutant) but R_kill = 0 under the PUT's primary MP. Six cells show R_sem $\geq$ 0.9 with R_kill = 0 (e.g. c1_CE1 GPR `noise_level` 1e-4 $\rightarrow$ 1e-1, c3_HP1 relu $\rightarrow$ tanh, d1_HP1 MLP α 1e-4 $\rightarrow$ 1.0; full list in Appendix C.4.1): the mutant is a valid injection, but the primary MP (partial-order for the c-class, monotonicity for the d-class) is an asymptotic or statistical relation whose tolerated parameter deviation absorbs the mutant. These examples also expose a family-boundary caveat: activation-swap or iteration-cap edits sit near the OS/HP and HP/TF boundaries, so the family label records generating intent, not a unique structural class, and the aligned-versus-cross reading carries this labelling slack.

S5 stratum-purity audit. We verified S5 directly rather than resting on generation intent. Running all five invariant checkers on each of the 292 admitted mutants (the deterministic offline AVP dispatcher, 20-repeat majority vote; SSOT `s5_purity_v4.json`) yields the invariant-flip distribution $\{0:170,\ 1:93,\ \geq 2:29\}$. The effect map σ is therefore single-valued on 263/292 (90.1%) of the corpus; the 29 multi-stratum mutants (9.9%) are confined to two operator families that mutate shared upstream state, control-flow reversal (CF, 9/9) and training-data corruption (TF, 20/54), while the four local-edit families (CE, OS, HP, SI) are 100% pure (0/229; Table 16). Re-attributing the aligned-versus-cross off-diagonal kill mass, 57/88 (64.8%) comes from pure single-invariant mutants (genuine cross-stratum detection) and 31/88 (35.2%) from multi-stratum mutants, the latter confined to the five cells `B2_MP1`, `C1_MP1`, `C1_MP2`, `D1_MP5`, and `D3_MP5`. The fiber partition thus survives at the σ level (R2's $\geq$ 90%-purity decision rule is met), and the aligned-versus-cross contrast should be read as a 57:31 pure/multi-stratum split rather than as uniformly pure cross-stratum detection.

Table 16. S5 stratum-purity audit of the v4 corpus. "Detected" = killed by at least one MP invariant checker; "pure" = killed by exactly one (S5 holds); "multi-stratum" = killed by two or more (σ multi-valued). Multi-stratum leakage is confined to CF and TF, which mutate shared upstream state.

Operator	Mutants	Detected	Pure (flip 1)	Multi-stratum ($\geq$ 2)
CE	64	27	27	0
OS	60	35	35	0
HP	72	11	11	0

SI	33	11	11	0
CF	9	9	0	9
TF	54	29	9	20
All	**292**	**122**	**93**	**29**

The cell-level evidence (Section 5.2) reproduces the pattern: 75% of cells are zero, concentrated in the cross slices, while v4 cross-source raises mean C1_share from 0.164 to 0.209 (+27%) among the few kills. Source diversity therefore improves the *quality* of the kill set without expanding its *coverage*. **The engineering insight is that operator-MP alignment in MR design is necessary for strong SMS signals; merely enlarging the semantically feasible mutant pool dilutes the C1 proportion without lifting the kill rate.** This motivates the source-axis reading in Section 5.3: the v3 → v4 micro-shift of −0.009 estimates LLM source diversity under a fixed prompt, while MR-design adequacy at the operator-MP level governs the kill rate.

5.10 RQ4: Industrial Construct-Separation Evidence

The industrial arm answers the last clause of RQ4: whether the construct separation among aggregate kill-rate, semantic alignment, and real-defect detection remains visible outside synthetic mutants and outside author-written kernels. The arm is a 34-case study over reproduced, MR-detectable defects from widely used scientific-computing libraries, drawn from an archived dataset [39] (10.5281/zenodo.21203424); each case pairs a buggy and a fixed revision and registers the metamorphic relation that discriminates the two. We report verified result totals only, and the arm does not change the paper into a benchmark article: defect mining, rejected cases, repository tooling, and benchmark release are left outside the paper and remain contributions of the dataset. The result-level evidence summary used here is included in the supplementary evidence package: it records the five-stratum counts, the aggregate mutation-phase statistics, the 34/34 real-defect face, and the non-nesting case identifiers, but not candidate mining or tooling. In the four-group mutation-phase comparison pre-registered in the dataset protocol, the four groups are the case's registered pattern-derived relation (T1), literature-generic baseline relations (B1), seeded random relations (B2), and mechanical ablations of T1 (A1); the comparison family is three one-sided case-level Wilcoxon signed-rank comparisons (T1 > B1, T1 > A1, B1 > B2) under Holm correction, and the case census (n = 34) was fixed by the dataset's verification status before any comparison was run. Under the primary all-mutant denominator the group kill rates are T1 377/1124 = 0.335 (Wilson 95% CI [0.308, 0.364]), A1 348/1124 = 0.310, B1 274/1124 = 0.244, and B2 228/1124 = 0.203. The pattern-derived group beats the literature-generic baselines (mean paired difference +0.101, bootstrap 95% CI [+0.029, +0.179], Holm-adjusted $p = 0.046$, Cliff's $\delta = +0.247$; one-sided Wilcoxon signed-rank $V = 279.5$, $z = 2.16$, unadjusted $p = 0.015$; case-resampling bootstrap $B = 10{,}000$, seed 20260704). Because the Holm-adjusted p sits close to the 0.05 boundary, we additionally report distribution-free checks of the *same* one-sided $T1 > B1$ contrast (robustness of the pre-registered estimand, not new hypotheses): an exact sign-flip permutation test on the signed-rank statistic (full 2^{27} enumeration over the 27 nonzero paired differences) gives $p = 0.014$, a Monte Carlo sign-flip test on the mean paired difference ($B = 10{,}000$) gives $p = 0.005$, and the BCa bootstrap 95% CI for Cliff's δ ($B = 10{,}000$) is $[+0.07, +0.46]$, excluding zero. The verdict is marginal under the Holm-adjusted normal approximation but robust under exact inference, and is reported with its fragility: the paired difference narrowed when the dataset grew from 30 to 34 cases, and a sensitivity rerun recorded in the dataset report (excluding the one case with a known tolerance-storm pathology) leaves the

Holm-adjusted verdict unchanged. The other two family members are not significant after Holm correction: the pattern-derived group does not significantly dominate its ablations, and the generic baselines do not significantly dominate the random draws. On the real-defect face, however, the distinction becomes sharper: the pattern-derived relation detects every real defect in the reported mutation-suite face (34/34). Because case admission itself requires an MR-detectable behavioural difference under the registered pattern-derived relation, this 34/34 count is selection-conditioned rather than evidential; the informative content of the face lies in the contrast on the same admitted cases: the literature-generic baselines detect nothing in 27 of 34 cases, the seeded random relations in 26 of 34, and the dimension-reduction ablation loses the real defect in 19 of 34 cases that its parent relation detects; a second, de-strictification ablation loses it in 17 of 34, with 11 cases lost by both (supplementary Appendix I).

The arm's cases were selected by a fixed inclusion rule: a case had to have a verified semantic fault, an executable pre/post or faithful contrast, and an MR-detectable behavioral difference under the declared stratum. Cases requiring unsupported build stacks, lacking an executable contrast, or failing MR-detectability verification were excluded before the reported mutation-phase comparison. No case was dropped after observing the comparative SMS or baseline detection results. The face is therefore a selection-conditioned, face-level check on construct separation, not a corpus-level claim about real-defect coverage.

This pattern is consistent with the construct separation predicted by the fiber view of Section 3.3.9. A mutant kill-rate is evidence about which sampled semantic effects an MR observes; it is not, by itself, a verdict on MR quality. The same industrial arm gives concrete non-nesting counterexamples, A-LAPACK-004, A-OPENBLAS-001, B-POCKETFFT-002, and E-ORDINARYDIFFEQ-001, where generic baselines kill mutants that the pattern-derived relation can miss or match. The mechanisms are distinct across the four cases, value destruction that preserves element order, zeroing that preserves a symmetry the relation checks, overshoot that stays within the tested bounds, and a variable-substitution/stale-read edit that operator mutation does not reach well, so no single relation family induces a total order over the others (supplementary Appendix I). Thus semantic mutation should be read as a certificate-bearing diagnostic for a declared semantic stratum, not as a scalar hierarchy in which one mutation family simply dominates another. This evidence therefore strengthens the paper's main argument while keeping benchmark construction outside the manuscript's contribution boundary.

5.11 RQ5: Cross-Class Consistency

Table 17. RQ5: Cross-class consistency and Friedman statistics.

Class	Mean SMS (v3)	Mean SMS (v4)
a (numeric)	0.067	0.067
b (probabilistic)	0.156	0.148
c (surrogate)	0.047	**0.089 (+91.5%)**
d (ML)	0.081	0.112 (+38.3%)

Sign test (within-class aligned mean − cross mean, sign = +): **3 / 4 (partial) under both v3 (same-source) and v4 (cross-source)**; the b-class is the inverted cell.

H3 verdict: not met. H3 carries two pre-registered criteria, a within-class sign test of 4/4 and CV(ΔSMS) < 0.5. The sign test is 3/4 (partial) under both v3 and v4, cross-source pooling does not flip the b-class inversion, so the consistency criterion fails. The per-class effects ΔSMS_c (aligned mean − cross mean) span from negative in the inverted class to large positive elsewhere, giving CV(ΔSMS) well above 1, far exceeding the 0.5 dispersion threshold, so the dispersion criterion also fails.

The mixed-effects primary model returned a Singular matrix, and its reduced fallback was degenerate (the full identifiability analysis is in Section 8.1); we therefore use Friedman as the non-parametric formal alternative:

- **PUT (n=12) × MP (k=5): chi^2 = 16.76, p = 0.0022** (significant; exploratory inferential per Table 10; v4 pool, `rq3_friedman_v4.json`).
- MP rank means: 3.88, 3.38, 2.38, 2.38, 3.00.
- Per-class Friedman with **Bonferroni × 4 correction**: a 1.000 / b **0.140** / c 0.924 / d 1.000, no per-class result remains significant after correction. Kendall's W effect sizes: a 0.333 (small) / b 0.862 (large concordance, but caveat: N=3 per class makes this label nominal only) / c 0.467 / d 0.417.

Caveat. Friedman tests "are MP rank differences present" (averaged over PUTs); H3 tests "is direction consistent across 4 classes". These are logically independent. H3 stands on the Section 5.11 sign test; per-class Friedman is descriptive only (small N=3 per class).

5.12 RQ5: Semantic-Mutation Score and Pattern Coverage

Status. The SMS–Pattern-Coverage relationship, a descriptive sub-question of RQ5, is reported as a descriptive observation, not a hypothesis test, because n = 12 places the 95% Spearman CI at roughly [−0.5, +0.6], so the test cannot distinguish zero, moderate-positive, or moderate-negative correlation. The numbers below are recorded so that a future study (n ≥ 30 PUTs) can pre-register a directional hypothesis.

Pattern Coverage (PC) per PUT = #triggered (MP_k, R_outcome) cells / 10. Range [0.500, 1.000], mean 0.733. Pairing with mean SMS over 5 MPs: Spearman rho = **0.163**; Kendall tau = 0.136; n = 12 (descriptive; no inferential reading attaches).

Statistical-power caveat first. At n = 12, Spearman's 95% CI is approximately [-0.5, +0.6] (Fisher-z approximation, a rough small-sample heuristic); the test cannot distinguish "zero correlation" from "moderate positive correlation" or "moderate negative correlation". Consistent with the descriptive licence of this row (Table 10, "none"), we attach no p-value; the descriptive ρ and τ do **not** support the strong claim that "SMS is independent of PC" or the strong claim that "SMS is strongly correlated with PC". The conservative finding is *no detectable correlation at n = 12*; orthogonality is a hypothesis for a future study (n ≥ 30 PUTs or refined PC operationalisation; full PC operationalisation in Appendix D.5).

5.13 Class-Level Stability and Source-Axis Interpretation

Section 5.11 gives the formal H3 verdict, so this deployment-facing reading is deliberately shorter: cross-source pooling improves the c- and d-class means (+91.5% and +38.3%), consistent with c / d classes having higher mutant-diversity demand than a / b, but it does not repair the b-class inversion or the CV(ΔSMS) dispersion failure. The mixed-effects singularity and the Friedman fallback are treated once (Section 8.1), not as a second independent result here.

5.14 Practical Interpretation and Deployment Scope

Scope. All deployment claims in this subsection are bounded to single-output `float` → `float` kernels under 2 KB, the 12 PUTs of this paper. They do not apply to industrial-scale, multi-module, or multi-output scientific computing software, which lies beyond the scope of this paper. Table 18 gives interpretive readings for SMS and LRCA outputs; it is not an acceptance-threshold table.

Table 18. Reading SMS and LRCA together.

Observation	Interpretation	Design response
Low SMS, high C1_share	The MR family detects few mutants, but the kills it does make are mostly semantic	Add MRs for adjacent semantic strata before enlarging the mutant pool
Low SMS, high suspect_share	The MR family mainly triggers tolerance, OOD, statistical, or artefact labels	Inspect tolerances, input domain, and statistical assumptions before treating kills as adequacy evidence
High SMS, low real-defect detection	Mutant kill-rate and defect detection have diverged	Revisit the declared semantic stratum and add real-defect-facing MRs
Generic relation kills outside the pattern-derived relation	Relation families are non-nested	Treat the result as a different semantic-effect fiber, not as domination by one MR family

Three stakeholder readings follow, with full deployment detail in Appendix E. **Test engineers** read SMS as a per-MR scalar: when it sits well below the aligned baseline of about 0.21 (Section 5.3), the LRCA labels (C2 tolerance, C3 OOD, C4 statistical) point to specific repair paths. **MR designers** use offline batch SMS as design feedback, on a quarterly audit (about 0.5 person-day per quarter; Appendix E.2) rather than per-pull-request gating, which LLM API latency, cost non-determinism, and air-gap incompatibility rule out. **V&V documentation** can carry SMS as research-grade supplementary evidence only: we make no normative IEC/ISO/ASME claim, and an earlier draft's acceptance thresholds were removed in revision as having no normative backing (Appendix E.3). Air-gap incompatibility is a hard limitation, the workflow needs external LLM API calls and is incompatible with most regulated air-gapped V&V environments (IEC 60880, DO-178C, IEC 62304, ISO 26262); pre-generated pools replay offline but new pools require LLM access (Appendix E.1). All three readings consume one source of truth (`paper_numbers_v4.json`, `lrca_60cell_v4.json`).

Example MR-design use. If a surrogate-model PUT has R_sem ≈ 1 for a Hyperparameter mutant but SMS = 0 under the primary monotonicity MR, the design action is not to generate more mutants of the same kind. The SMS / LRCA reading is that the MR family is blind to the admitted semantic-effect fiber; the next design step is to add a convergence, stability, or fidelity-order MR for that PUT before claiming adequacy for the hyperparameter stratum.

6 Study 2: A Pre-registered Confirmatory Audit at $n = 28$

The 12-PUT audit above (Study 1) delimits the semantic-mutation construct through four threshold misses, H1–H4. Those misses are honest, but they are also *unpowered* in ways that make them weak evidence either for or against the construct. The operator-implementability bar assumed uniform applicability (≥ 9/12 PUTs for every family), which class-targeted operators cannot meet by design. The aligned-versus-cross verdict rests on a point-estimate rule whose pass probability sits near one half when the truth is at the Romano boundary (Section 5.4). The cross-class sign test is a 4/4 criterion at $n = 4$ classes with heavy ties. The attribution bar averages `suspect_share` over 60 non-independent cells. Study 1 also carried two further temptations that a confirmatory design must foreclose: a data-dependent primary-meta-pattern choice (the prohibited v3b path) and a protocol asymmetry between the same-source and cross-source arms. Study 2 re-registers powered or feasibility-bounded successors to these four questions, fixes the primary-meta-pattern rule deterministically before any generation, and executes the campaign once under a frozen registration. It is a confirmatory study, not a second exploratory pass; everything discovered after the freeze is exploratory by definition.

6.1 Design and registration

The registration of record is `docs/prereg_v2/PREREGISTRATION_STUDY2_v1.1.md`, frozen with a pre-data attestation before any Study-2 mutant, review label, or SMS cell existed (it amends an earlier un-minted v1.0 and strengthens its integrity machinery). The freeze is self-attested through repository history: the pre-data commit hashes recorded in the registration and `PILOT_LOG.md` fix its precedence over data generation; external timestamping (a Zenodo mint) is pending, so a reviewer verifies freeze-precedence against the repository rather than a third-party registry. The design has four fixed elements.

- **Blind-authored 30-PUT pool.** An 18-PUT expansion (roster in `PUT_EXPANSION_ROSTER.md`) was authored from domain knowledge only, with no mutation-outcome file read during authoring, and added to the original 12. A 2-PUT calibration pilot $\{a2, b4\}$ is carved out for pipeline debugging with outcomes visible, and is *excluded from every confirmatory analysis*; the confirmatory grid is therefore **28 PUTs** at class balance A7/B6/C7/D8.
- **Powered successor hypotheses.** H2-1′ (aligned slice dominates cross, one-sided Cliff's $\delta > 0$, power 0.9285 at $n = 28$); H1′ (≥ 4 of 5 operator families produce ≥ 5 non-equivalent mutants on ≥ 8 of 28 PUTs, feasibility 0.843); H3′ (positive aligned>cross direction in $\geq$ 3 of 4 classes, power 0.949); H4′ (mean `suspect_share` ≤ 0.05 over the 140 confirmatory cells, tightened from Study 1's 0.20 under a single-stratum spec constraint). All thresholds trace to the power SSOT `power_study2_v11.json` (seed 20260708).
- **Deterministic primary-meta-pattern rule.** The class rule (A→MP1, B→MP2, C→MP5-held, D→MP2) is fixed at authoring, blind to mutation results; the Study-1 v3b selection-on-the-response path is prohibited. The rule is stated in the registered MPk labels under which it was authored; in NOETHER's names this is A→ m_{inv}, B→ m_{mono}, C→ m_{cmp}-held, D→ m_{mono}.
- **One-shot rule.** Confirmatory generation runs once per the registered budget with seeds and prompt-template hashes pinned; regeneration, cell cherry-picking, or moving any threshold after a confirmatory outcome is visible is a protocol violation to be reported as such.

6.2 Harness instantiation and the gated cross-vendor question

We disclose the harness plainly because it bounds one hypothesis. Study-2 generation and review run through the Claude Code Agent harness: an Opus-tier generator role, separately spawned and role-isolated Claude reviewer instances receiving *blinded* packets (no generator identity, arm label, or kill outcome), and a third Claude instance for arbitration. Role isolation and packet blinding are enforced, but all instances share one model family, a **same-vendor** instantiation. It can test the same-source questions H2-1′, H1′, H3′, H4′, but *cannot* instantiate the cross-vendor dual-blind contrast, so H2-2 is **registered but gated: reported not-run, with no same-vendor proxy substituted**, left open rather than reported as a null (resolved model id recorded in the campaign SSOT). The v5 pools carry per-mutant source tokens `claude`, `gpt`, `deepseek` inherited from Study 1's three-slot layout; under this harness these are nominal *slot labels*, all served by the same Claude-family generator, and the aligned-versus-cross axis is defined by the meta-pattern match, not the slot, so no confirmatory statistic partitions by vendor slot.

6.3 Execution

The confirmatory campaign ran once. Generation produced 774 valid mutants across the 28 PUTs; mechanical admission gates V1–V4 admitted 756, giving pools of 756 mutants over 28 PUTs and 140 confirmatory cells (28 PUTs × 5 meta-patterns), of which 36 carry nonzero killed mass. Blind review produced 747 verdicts from 18 independent reviewer instances; 4 mutants were rejected at the first round, and 6 uncertain cases went to third-instance arbitration, all 6 resolved as confirmed under a single uniform rationale (pinning the Gaussian-process hyperparameters before comparison). SMS was computed only after the review labels were frozen.

Three items are disclosed from the confirmatory phase (full log in Appendix J). *Incident P6*: the scoring tool defaulted to the Study-1 track and overwrote `sms_track1.json`; the file was restored from the tracked tree before any use and the run repeated with an explicit track-2 target, a deterministic re-execution, not regeneration. *Incident P7*: a blinding assertion fired when one PUT's mutants carried a source-family token in their docstrings; a presentation-layer redaction was added at export (displayed code only; on-disk artefacts untouched; assertion retained) and the packets re-exported. *Deviation D-A1*: the pre-frozen δ script gained a post-data mode that skips only the gated H2-2 computation and emits its registered not-run verdict; the H2-1′ logic is byte-unchanged, and the deviation is recorded here and in the output SSOT.

6.4 Results: the confirmatory scoreboard

We lead with the full scoreboard, including the miss. Three of the four executable successor hypotheses confirm; the attribution-purity hypothesis H4′ does *not*; the cross-vendor hypothesis H2-2 is gated not-run. The five verdicts (H2-1′ CONFIRM, H1′ CONFIRM, H3′ CONFIRM, H4′ NOT_CONFIRMED, H2-2 NOT-RUN) and their permissions are the Study-2 rows of the cross-study ledger (Table 10); the registered thresholds and key statistics are reported per hypothesis in the paragraphs below.

H2-1′ (confirm). On the 28 aligned versus 112 cross primary-cell SMS values, Cliff's $\delta = +0.4295$ with a one-sided 95% percentile-bootstrap lower bound of +0.2653 ($B = 10{,}000$, seed 20260708), clearing the registered > 0 rule. The licensed claim is *directional*: the aligned slice stochastically dominates the cross slice. The two-sided 95% CI $[0.2328, 0.6193]$ and the descriptive Romano medium band (≥ 0.330) are reported for context only and are never promoted to a large-effect pass; the study deliberately does not re-assert Study 1's 0.474 large-effect bar.

H1′ (confirm). All five families clear the $\geq 8/28$ bar: CE on 23, OS on 14, HP on 21, TF on 15, and SI on 8 PUTs (against outcome-independent coverage ceilings of 23/14/21/15/10). Notably SI, which the registration expected to remain below the bar (feasibility 0.843, not a rubber stamp), clears at exactly 8, so the pass is $5/5$, stronger than the $4/5$ required. This converts Study 1's non-uniform applicability finding into a powered adequacy result on the wider grid.

H3′ (confirm, with one negative class). Class-mean aligned minus cross SMS is positive in three of four classes: A (0.3333 vs 0.0119), B (0.2222 vs 0.0903), and D (0.3333 vs 0.0937); class C is *negative* (0.0476 vs 0.0714) and is reported as such. The descriptive per-class binomial sign tests (under-powered, reported only for context) give one-sided p of 0.0312 (A), 0.1875 (B), 0.9688 (C), and 0.0352 (D). The across-meta-pattern Friedman test ($\chi^2 = 26.23$ over 28 blocks and 5 treatments) stays exploratory, exactly as in Study 1.

H4′ (not confirmed). The mean suspect_share over the 140 confirmatory cells is 0.1714, above the registered 0.05 bar and far above the projected rule-of-three upper bound of 0.0131. The per-family multi-stratum breakdown locates the leakage: CE (0/207) and HP (0/189) stay clean, but TF (72/135) and OS (27/123) leak heavily, with CF (9/18) and SI (9/69) contributing the remainder. We must disclose a design-intent defect that bears on how this number should be read (incident P8, supplementary Appendix J.4): the registered single-stratum admission screen for CF and TF was wired but *ineffective*. Because the cross-source pool builder tagged each candidate with a model-source suffix (for example c7_TF1_claude), the screen's category regex failed to match, returned an unconstrained category, and admitted every candidate without ever evaluating its flip count; a post-hoc check confirms all 81 CF/TF multi-stratum mutants were admitted this way, and the pilot's apparent "9/9 single-stratum pass" (PILOT_LOG.md) was the same false negative. H4′ therefore evaluated the *unscreened* pool. The registered decision rule scores suspect_share on the admitted pool and does not condition on the screen, so the frozen NOT_CONFIRMED verdict is unaffected in validity; what was not realised is the design *intent* that CF/TF enter already single-stratum. OS and SI were never in the constrained set by design, so the screen never targeted them, and OS, which showed no incidental multi-stratum leakage in Study 1, now leaks on the richer grid. We do not fix the screen here; correcting it belongs to the Study-3 registration wave.

6.5 Interpretation

The three confirms turn Study 1's boundary-delimiting misses into powered, positive statements about the same construct. Operators instantiate adequately on a blind-authored grid (H1′); the aligned slice dominates the cross slice as a *directional* claim at medium descriptive magnitude (H2-1′); and the aligned>cross direction holds across most design classes (H3′). The claim ladder is precise: Study 1 delimits the construct, and Study 2 confirms the delimited construct *directionally*. We do not upgrade the medium descriptive band into a large-effect claim, and we do not read the single negative class (C, surrogate models) as noise: it is a registered-a-priori signal that cross-class consistency, though now confirmed at the $\geq 3/4$ threshold, is not uniform.

The H4′ miss is a genuine finding, not a nuisance. In Study 1 the multi-stratum leakage was confined to the CF and TF families; the registration projected that a single-stratum spec constraint would push residual leakage down to the clean-family regime. That constraint never actually filtered (incident P8): H4′ measured the unscreened pool, in which attribution leakage *generalises* beyond the CF/TF diagnosis, with TF still leaking the most (72/135) and OS leaking newly (27/123). The post-hoc diagnosis in Section 6.6 shows this leakage is construct-level and reproducible rather than a scoring artifact, so LRCA-based single-stratum attribution remains an open construct boundary rather than a solved problem. Closing it is an open problem, not a promised third study: it would require either a per-family single-stratum re-specification validated on the widened grid (so that the scoring filter and the admission filter agree cell by cell), or a richer stratum model that admits declared multi-stratum operators and scores them accordingly. Until then, the honest reading is that aligned>cross magnitude and direction survive a powered confirmatory test, while clean semantic-effect attribution does not.

6.6 Post-hoc diagnosis of the H4′ leakage

This subsection is **post-hoc and exploratory**. It decomposes the H4′ leakage mechanistically; it does not re-litigate the registered 0.05 threshold or re-register a hypothesis. The frozen verdict remains NOT_CONFIRMED (mean suspect_share 0.1714). Every number below is emitted by scripts/diagnose_h4_leakage.py into the SSOT h4_leakage_diagnosis_v5.json from committed artifacts only.

The leakage is construct-level, not a measurement artifact. Splitting the 117 multi-stratum confirmatory mutants (CF 9, OS 27, SI 9, TF 72) by whether they stay multi-stratum under a stronger repeats = 20 re-measurement gives **B** = 117/117**, A** = 0: every flagged mutant reproduces its exact flipped-invariant set, so the leakage is a property of the mutants, not of the single-shot scoring pass. Each family-by-class cell collapses onto a single co-flip pair (a deterministic coupling, not scatter; per-cell fingerprints in supplementary Appendix J.4), dominated by the $[m_{\text{mono}}, m_{\text{cmp}}]$ [MP2, MP5] coupling: TF's 72/135 kills at m_{mono} and at m_{cmp} are the *same* 72 mutants.

The driver is PUT richness, not measurement. Because A = 0, the repeats = 20 → 1 gap cannot explain leakage growth from Study 1 (CF 9, TF 20; OS, SI 0) to Study 2 (TF 72, OS 27, SI 9). Class-c surrogates and class-d classifiers *train on data*, so a training-data corruption (TF) perturbs both monotonicity and asymptotic ordering at once, and even local-edit operators (OS, SI) straddle invariant classes on these richer subjects; CE and HP stay clean throughout, fixing the phenomenon to rich PUTs, not any operator alone. As a mechanical what-if (not a re-verdict), a correctly wired all-family screen would reject all 117 and drive counterfactual mean `suspect_share` to 0.0, but that trades attribution purity for a selection-biased pool, removing multi-invariant structure that should be modelled, not noise to filter.

Interpretation. The reproducible B = 117/117 split indicates that on structurally rich (surrogate and ML) PUTs a single-edit fault *inherently* perturbs several invariants, so strict single-stratum purity and a single-valued σ are the wrong model there. A graded, multi-valued attribution construct is the registered path taken by Study 3 (Section 7), which characterises this boundary on both sides rather than rescuing the miss.

An MR-sensitivity coverage boundary. The same artifacts expose an orthogonal boundary: convergence order (MP3, m_{conv}) records zero nonzero detection cells in both studies (Study-1 v4: 0; Study-2 v5: 0 of 28) and trajectory determinism (MP4, m_{dyn}) is near-zero (0; 1 of 28), and this is not designed-vacant coverage (m_{conv} is exercised on 26 of 28 subjects yet detects nothing). The two strata mark an MR-sensitivity boundary, declared and instantiated but not empirically stressed by the current operators and AVP checkers, distinct from the attribution-purity boundary above.

7 Study 3: A Pre-registered Graded-Attribution Follow-up

Study 2's attribution verdict (H4′, NOT_CONFIRMED) and its post-hoc diagnosis (Section 6.6) left one question sharply posed but unsettled. The diagnosis showed the multi-stratum leakage is construct-level and reproducible (bucket split B = 117/117), not a scoring artifact, and it exposed a design-intent defect (incident P8): the single-stratum admission screen was a silent no-op for the whole v5 campaign, so H4′ scored an unscreened pool. Two consequences followed. A single-valued effect map σ is the wrong model where a single fault inherently perturbs several strata, so the purity bar tested the wrong construct on rich PUTs; and the screen itself had to be repaired and its effect *verified in production* before any purity claim could be trusted anywhere. Study 3 re-registers the attribution construct alone to settle both, on fresh data, without re-opening any other Study-2 verdict.

7.1 Design and registration

The registration of record is `docs/prereg_v2/PREREGISTRATION_STUDY3_v2.md` (v2.0), frozen with a pre-data attestation before any Study-3 mutant, cell, or verdict existed; it supersedes only the frozen Study-2 H4′ and leaves every other Study-2 verdict untouched. Study 2's v5 data is used for a single purpose, calibration of the graded data-generating process and thresholds (power SSOT `power_study3.json`, seed 20260708), stated openly. The design has four fixed elements.

- **A split estimand.** The single purity bar is replaced by two confirmatory verdicts in one Holm(2) family (Family G). *H4″-graded* (primary) asks whether the declared MetaPattern still carries substantial *graded* attribution on the structurally rich classes: for each detected mutant m declared to primary stratum m^*, $s_m = \mathbb{1}[m^* \in \text{flipset}(m)]/|\text{flipset}(m)|$, aggregated as the rich-class (C, D) mean, with a registered bar of mean share ≥ 0.15 (power 0.82 at $n_{\text{rich}} = 15$). *H4″-strict* asks whether single-stratum purity holds where coupling is absent or screenable, over {CE, HP, CF-with-screen}, with a bar of purity ≥ 0.90 (one-sided 95% lower Clopper–Pearson). The declared strata are the frozen class-primaries, in NOETHER's names C→ m_{cmp} and D→ m_{mono} (registered labels MP5-held and MP2); the graded measure *reports* rather than repairs the C→MP5 mismatch.
- **A repaired, live screen.** The P8 defect is fixed forward for the Study-3 path only, altering no committed v5 artifact: the op-id parser tolerates the model-source suffix, the screen covers all families (`P2_SCREEN_ALL_FAMILIES = 1`), and a null category raises loudly instead of admitting silently. A registered screen-smoke gate makes the campaign fail loudly if the wired screen matches zero candidates, so the P8 no-op can never recur undetected.
- **Fresh generation, one shot.** A hypothesis about attribution formed after seeing v5 cannot be confirmed on v5, so Study 3 regenerates all mutants with fresh seeds into a fresh validated pool (SSOT `sms_track2_v6.json`); the confirmatory grid is the same blind-authored 28 PUTs, and generation runs once per the registered budget. The $\{a2, b4\}$ calibration pilot is reused verbatim and excluded from every confirmatory statistic; the only pilot-triggered change was code-level v6 tooling wiring (defect P9), which touches no threshold, estimand, or roster.
- **Multiplicity restraint.** The settled Study-2 successors H1′, H2-1′, H3′ are *not* re-registered; re-testing them on fresh data would inflate family multiplicity without posing a new question. Study 3 registers exactly the two attribution verdicts plus the screen-smoke assertion.

7.2 Execution

The confirmatory campaign ran once through the same same-vendor Claude harness as Study 2. Generation produced 765 valid records; V1–V4 gates, now including the live all-family screen, admitted 720 and rejected 45 (a3, b3, c1, d8). Pool assembly removed a further 87 to give v6 pools of 633 over the 28 PUTs (678 with the $\{a2, b4\}$ pilot): 81 fell to the single-stratum clean-attribution screen (nine PUTs each shedding one multi-stratum family of nine, e.g. d8 18 → 9), and 6 to the per-PUT cap of 30 (a1 36 → 30). This yields 140 cells, of which 20 carry nonzero killed mass. Blind review returned 720 verdicts with zero schema errors and zero arbitrations, and SMS was computed only after labels were frozen.

The screen now does real work. On the confirmatory pool the all-family re-audit matched all 633 mutants and flagged 36 as multi-stratum, where the byte-identical v5 screen had matched *zero* candidates (the P8 silent no-op); the registered smoke gate therefore passes loudly rather than masquerading as an unconstrained pass. The P8 remediation is thus verified in production, not merely in a unit test. The single post-freeze code change is the v6 tooling wiring (defect P9, supplementary Appendix J.6); the pre-frozen scorer `compute_h4_graded.py` produced both verdicts without modification.

7.3 Results: a two-sided attribution boundary

Both registered verdicts are front-loaded. Attribution is now *positively validated* where coupling is absent or screenable, and *positively refuted* on the rich classes; the two together characterise a boundary rather than rescue the Study-2 miss. The two Family-G verdicts (H4″-strict CONFIRM, H4″-graded NOT_CONFIRMED) and their permissions are the

Study-3 rows of the cross-study ledger (Table 10); the registered thresholds and key statistics are given per hypothesis below.

H4″-strict (confirm). Over the 90 detected clean-family mutants (CE 72, HP 18; no CF mutant was detected on this pool), every one is single-stratum, so purity is 1.0 with a one-sided 95% lower Clopper–Pearson bound of 0.9673, clearing the registered 0.90 bar; the screen matched > 0 candidates, so the smoke gate does not force a loud fail. The licensed claim is exactly the registered one: the single-valued σ model is valid for {CE, HP, CF-with-screen}, and is *not* claimed for OS, SI, or TF on rich PUTs.

H4″-graded (not confirmed). The rich-class mean primary-stratum kill share is 0.0833, with a one-sided 95% percentile-bootstrap lower bound of 0.0 ($B = 10{,}000$, seed 20260708), below the registered 0.15 bar; the point estimate itself misses the bar, so the verdict does not rest on power alone. The achieved $n_{\text{rich}} = 6$ (the rich-class PUTs that produced any detected mutant: C3–C6, D2, D5) is below the registered target of 15 because most rich cells detect nothing, and it is reported as such. The per-class breakdown is stark: class C (declared m_{cmp}) has mean share 0.0 across its four contributing PUTs, and class D (declared m_{mono}) has 0.25 across two. Mechanistically, the detected rich-class kills fire on a MetaPattern *other than* the declared stratum: on C, the CE and TF kills flip conservation (m_{inv}), not method-comparison; on D, the OS kills flip m_{inv} and only the SI co-flip touches the declared m_{mono}, as half of an $\{m_{\text{mono}}, m_{\text{cmp}}\}$ pair. The declared stratum is not where the rich-class signal lands.

7.4 Interpretation

Study 3 turns the Study-2 attribution miss into a two-sided characterisation. Where a fault touches a single stratum or a stable coupling can be screened out deterministically (CE, HP, CF), single-stratum attribution is now *positively validated*, with the screen verified live; on the structurally rich classes (surrogate and ML), declared-stratum attribution is *positively refuted* even in graded form, because a single fault perturbs several invariants at once and the kill signal concentrates off the declared MetaPattern. This sharpens rather than reverses Study 2: the repaired screen changes the clean-family picture (purity 1.0) without touching the rich-class refutation, confirming the earlier failure was construct shape, not a wiring artifact. The claim ladder is now three rungs (Study 1 delimits, Study 2 confirms directionally, Study 3 bounds the attribution boundary). We do not read the rich-class refutation as a defect to patch away: a screen forcing those mutants single-stratum would discard the multi-invariant structure trained models exhibit. Whether a richer multi-valued attribution model can score that structure is left open; Study 3 registers no fourth study.

8 Threats to Validity

Table 19 summarises the threat areas by class; the complete per-threat table, including all Study-2 and Study-3 harness, screen-remediation, and attribution disclosures, is reproduced in full in supplementary Appendix K.1. No threat is dropped: the summary rows below name the area, and every mitigation and disclosure is carried verbatim in the appendix table.

Final limitations. We list ten known limitations.

(1) **Equivalence determination is a probabilistic approximation.** $K_{\text{eq}} = 1000$ sampling implements the undecidable equivalence problem; a Hoeffding false-equivalence bound is in Appendix F.1, but the $K_{\text{eq}} \in \{500, 1000, 2000\}$ sweep is deferred to a follow-up.

Table 19. Threats to validity, summary by area (full per-threat table with all disclosures in supplementary Appendix K.1).

Threat area	Class	Mitigation / disclosure (detail in Appendix K.1, F)
Internal (reproducibility, infrastructure, pool, non-determinism, protocol asymmetry)	Internal	Archived raw responses and frozen pools; AVP pinned to upstream hash with embedded source; C1_share 0.164 → 0.209 shows quality not down under v3→v4 asymmetry (Appendix F.1)
Construct (E2 approximation, LRCA labels, LLM homogeneity, same-vendor Study-2 harness)	Construct	K_{eq} = 1000 + Hoeffding bound; 3-LLM rotation + 20% manual sampling; Study-2 same-vendor mono-culture disclosed with H2-2 gated not-run, not proxied (Sections 6.2, F)
External (12-PUT representativeness, LLM-source shift, HOM, real-defect arm)	External	Coverage argument (Appendix B.1); categorical (not frequency) HP/SI/TF argument; industrial arm used at result level only, benchmark release deferred (Section 5.10)
Attribution scoring/admission gap (Study 2)	Construct	H4′ leakage 0.1714 reported as a genuine miss; scoring and admission filters disagree cell-by-cell; closing it an open problem (Section 6.5)
Screen remediation and rich-class attribution (Study 3)	Internal / Construct	P8 fix verified in production (633 matched, 36 flagged vs zero under v5); graded rich-class attribution refuted (mean share 0.0833, n_{rich} = 6) (Sections 7.2, 7.4)

(2) **LLM generation homogeneity bias.** Training-data overlap across Claude, GPT, and DeepSeek may share blind spots; three-LLM rotation and 20% manual sampling reduce but do not eliminate it, and a fourth independently trained (self-hosted open-weight) family is the natural strengthening.

(3) **Limited power for cross-class consistency.** The four-class sign test has df = 3 and the mixed-effects model is singular at $N = 60$; H3 is exploratory, reported as class means, sign test, Friedman, and a forest plot.

(4) **LRCA reports "likely" root causes.** The priority C5 > C4 > C3 > C2 > C1 is an engineering choice; the package preserves the full multi-label co-occurrence table, so `root_cause` is a likely, not definitive, attribution.

(5) **The AVP is reused from shared infrastructure.** The package embeds the AVP source for self-consistency, but interface semantics may shift under a major upstream revision.

(6) **Epistemological vs engineering scope.** SMS measures semantic detection capability; engineering value is not evaluated here.

(7) **Signature simplification.** The `float` → `float` single-output signature is a substantive constraint bounding mutant semantic complexity; a portability study should validate SMS on multi-output PUTs.

(8) **Air-gap incompatibility.** The generation workflow calls external LLM APIs, incompatible with most regulated air-gapped V&V environments (IEC 60880, DO-178C, IEC 62304, ISO 26262); pre-generated pools replay offline but new pools require LLM access.

(9) **No proof-carrying semantic-effect certificates.** E1 ∧ E2, LRCA, and AST audit trails are empirical and structural certificates, not machine-checkable proof objects: we do not yet define an abstraction function to invariant domains or attach per-mutant proof/error-expression certificates binding an edit to a declared semantic-effect class. This is the main gap to a certificate-bearing framework.

(10) **Real-defect evidence is used narrowly.** The industrial arm shows kill-rate, alignment, and real-defect detection separating on reproduced library defects; benchmark construction, curation, tooling, and release remain contributions of the archived dataset, and the SMS instantiation itself is evaluated on the 12 controlled kernels only.

8.1 Statistical-Modelling Capacity

The mixed-effects primary model `sms ~ C(class) * C(operator) + (1 | put)` returned a Singular matrix at N = 60; the fallback `sms ~ C(class) + C(operator) + (1 | put)` had the PUT random-intercept variance hit boundary 0 (degenerate). Both failures are not numerical accidents; rather, they are evidence that 60 observations across 12 PUTs cannot identify an 11-dimensional fixed-effects structure plus 12-PUT random intercepts. This is a hard limit of the present design: **the experiment cannot, in principle, fit a hierarchical model with this fixed-effect dimensionality at this sample size**. Friedman tests serve as the non-parametric formal alternative, but they answer a structurally different question. Expansion to n_PUT ≥ 30 is the only path to a properly identified hierarchical model.

8.2 Programs with Zero Observed Signal

Figure 2 and Section 5.9 document that PUTs A1, A3, and C2 produce all-zero rows across the five MPs in the cross-source pool; that is, **for 25% of the PUTs, SMS gives zero signal at every operator-MP cell**. This is a substantive limitation of SMS as an adequacy metric on this PUT cohort: when the LRCA C1 mass collapses to zero across all aligned and cross slices for a PUT, SMS cannot distinguish strong from weak MR sets on that PUT. The phenomenon coexists with R_sem ≈ 1.0 (mutants are valid; Section 5.9), so the issue is not mutant generation but MR-design adequacy at the PUT level. MR-design refinement (per-PUT MP discovery) and the proposed pre-registered c-class primary-MP rule on a fresh dataset are the principal mitigation paths. We declare this as a residual PUT-level zero-signal threat.

8.3 Protocol-Asymmetry Magnitude Estimate

Section 4.9 declared protocol asymmetry: v3 used the Phase-1 dual-blind reviewer protocol (Claude generation + GPT-5.4 review + DeepSeek arbitration), while v4 passes V1-V4 mechanical gates only. To quantify the potential δ-shift attributable to this asymmetry without dual-blind, we offer a **rough order-of-magnitude estimate**: in the Phase-1 audit, dual-blind review filtered approximately 5-10% of LLM-generated mutants as semantically inconsistent; if these filtered mutants were systematically less effective at killing under MR_aligned (a plausible but unverified premise), the upper-bound δ-shift contribution from removing the reviewer step is bounded above by approximately **±0.03 to ±0.05 on Cliff's δ** at n_aligned = 12. This bound is an order of magnitude larger than the observed −0.009 source-axis shift and therefore prevents reading the source-axis contrast as evidence of LLM-source-diversity inertness; a direct dual-blind v4 rerun (deferred to a follow-up study) would give a tighter quantification. Such a rerun on the same 12 PUTs only removes the protocol-asymmetry confound: at $n = 12$ the $\Delta\delta$ CI stays ≈ 0.44 wide, so detecting a $\Delta\delta = 0.20$ effect at 80% power requires ≈ 30 PUTs, and the follow-up is planned at $n \geq 30$.

9 Future Work

Seven directions follow: (1) a pre-registered c-class primary-MP rule on a fresh dataset; (2) a differential-prompt LLM-diversity test (V_canonical, V_persona, V_cot; Appendix C.1.1); (3) a scaling study at $n \geq 30$ PUTs for SMS-vs-Pattern-Coverage orthogonality and HOM-equivalence empirics; (4) an abstract-semantics certificate layer defining the abstraction α to invariant domains and attaching per-mutant proof/error-expression certificates; (5) a dual-blind rerun on the full v4 grid to separate protocol asymmetry from source diversity; (6) cross-language portability (Python → JavaScript, Julia) building on Tip et al. [47]; and (7) a self-hosted open-weight LLM pilot for air-gapped deployment.

10 Conclusion

10.1 Findings summary

The empirical work supports the semantic-mutation construct in nine ways: the first seven from Study 1, the eighth from the Study-2 confirmatory audit, and the ninth from the Study-3 graded-attribution follow-up. The Study-1 pre-registered thresholds are treated as stress tests of this instantiation, not as prerequisites for the SMS definition.

(1) SMS preserves the classical MS ratio while moving mutant generation, equivalence, and killing into declared semantic strata, so the 60-cell audit evaluates how one MR design observes semantic-effect fibers rather than redefining mutation testing.

(2) The structural audit separates the semantic and syntactic mutant spaces: 5.14% AST-normalised overlap across 12 PUTs, with HP, SI, and TF at 0/72, 0/33, 0/54 under first-order defaults.

(3) Operator-MP alignment produces a positive but not large effect: the H2 large-effect threshold is **not met** under the point-estimate criterion ($\delta_{v3} = 0.323$, below the Romano et al. [45] 0.330 medium threshold), and the stipulated-alternative analysis shows the rule passes near one half at the boundary, so this is a point-estimate result, not a proof the effect is below large.

(4) Cross-source pooling changes mutant quality more than aligned-versus-cross separation: holding the c-class primary at partial-order, three-LLM pooling shifts δ by only -0.009 (95% CI $[-0.238, 0.207]$) while raising mean C1_share by 27%, class-c mean SMS by 91.5%, class-d by 38.3%.

(5) The MP structure is visible (Friedman $\chi^2 = 16.76$, $p = 0.0022$) but cross-class consistency H3 is **not met** (sign test 3/4, CV(ΔSMS) > 0.5); SMS-vs-Pattern-Coverage is 0.163 at $n = 12$, so orthogonality remains a future hypothesis.

(6) The industrial real-defect arm supports construct separation beyond the PUT audit: pattern-derived relations detect all 34 tabled real defects in the reported face, while the four-group comparison shows only a modest aggregate advantage and non-nesting cases persist, so kill-rate, semantic-stratum alignment, and real-defect detection are related but distinct constructs on industrial code.

(7) The remaining failed thresholds delimit the instantiation: H1 is not met (only HP reaches ≥ 5 mutants on $\geq 9/12$ PUTs) and H4 is not met (mean suspect_share 0.791); neither changes the degeneration theorem or the strong-boundary analysis.

(8) A second, pre-registered confirmatory study (Section 6) re-runs the four questions with powered successors on a blind-authored 28-PUT grid, executed once. Three confirm: H1′ (5/5 families clear $\geq 8/28$); H2-1′ ($\delta = +0.4295$, one-sided 95% lower $+0.2653$, directional, not large); and H3′ (3/4 classes, class C negative). Attribution purity H4′ is *not* confirmed (mean `suspect_share` 0.1714 vs a 0.05 bar): multi-stratum leakage generalises beyond the Study-1 CF/TF diagnosis (TF 72/135, OS 27/123) once the grid widens. The cross-vendor hypothesis is gated not-run under the same-vendor harness.

(9) A third, pre-registered study (Section 7) re-registers attribution alone on fresh mutants with the P8 screen repaired and verified live (633 matched, 36 flagged vs zero under v5), splitting it into two Holm-corrected verdicts: single-stratum purity is *confirmed* (H4″-strict: purity 1.0, 95% lower CP $0.9673 \geq 0.90$) on {CE, HP, CF-with-screen}, while graded declared-stratum attribution is *refuted* (H4″-graded: rich-class mean share 0.0833, lower $0.0 \leq 0.15$) on the rich surrogate and ML classes, where the kill signal fires off the declared MetaPattern. The three studies read as: Study 1 delimits the construct, Study 2 confirms it directionally, Study 3 characterises the attribution boundary on both sides.

10.2 Methodological contributions

This paper contributes a three-layer framework for domain-semantic mutation in single-output scientific computing kernels.

- **Layer 1 (Section 4.2).** Formal necessary conditions for semantic mutation, cross-function-boundary substitution, dependence on domain knowledge, change in algorithmic class, instantiated as five operator families: CE, OS, HP, TF, and SI (also written mut_C, mut_M, mut_G, mut_T, mut_F).
- **Layer 2 (Section 3.3.3).** The E1 ∧ E2 equivalence judgement, the conservative complete instantiation of the Layer-1 conditions, with an explicit trade-off against E1-alone and E2-alone variants.
- **Layer 3 (Section 4.5).** AST-normalised empirical traceability across all 12 PUTs: a 5.14% overall AST overlap with cosmic-ray defaults, and HP, SI, and TF unreachable at 0/0/0 under default first-order configurations (HOM not refuted; see Section 4.6).

SMS is backward-compatible with the classical MS through Theorem 3.1 (Section 3.3.6), which establishes almost-everywhere degeneration in the limit $L = L_{\text{equiv}} \wedge L_{\text{killed}} \wedge L_{\text{mut}}$. The 60-cell empirical audit reported in Section 5 is one demonstration following this backbone, not the paper's main contribution.

Data and Code Availability

All Study-1 raw data, JSON SSOTs (paper_numbers_v3/v4, rq2_cliffs_delta_v3/v4_mp5, rq3_friedman_v4, s5_purity_v4, h2_incidence_v4, lrca_60cell, lrca_v4_mp5_recompute, rq2_power_stipulated, cosmic_ray_12put_ast_diff), mutant pools, AVP source, and analysis scripts are archived on Zenodo under DOI 10.5281/zenodo.20250664. The industrial arm (Section 5.10) draws its result-level statistics from a separately archived defect dataset [39] (10.5281/zenodo.21203424; benchmark construction is that dataset's contribution, not this paper's); every RQ4 number is recomputable in-repo from the per-case SSOT `data/results/industrial_percase_v1.json` (provenance DOI 10.5281/zenodo.21203424, archive SHA-256 recorded), via `scripts/build_industrial_ssot.py`, which fails the build on any mismatch. The Study-2 audit is backed by the frozen registration `PREREGISTRATION_STUDY2_v1.1.md` and `PILOT_LOG.md` (incidents P1–P7, deviation D-A1), with every Study-2 number tracing to a pre-frozen SSOT (`sms_track2_v5.json`, `dualblind_delta_delta_v5.json`, `h1_instantiability_v5.json`, `h3_class_consistency_v5.json`, `s5_purity_v5.json`; powers in `power_study2_v11.json`, seed 20260708). An earlier version is on arXiv: arXiv:2605.17437 [cs.SE]. A read-only repository mirror is available to the Editor on request; structure follows `REPRODUCIBILITY.md`, and the cosmic-ray/mutmut operator versions are pinned in `requirements-frozen.txt`.

Supplementary Material

Appendices A–I (notation and operator catalogue; experimental subjects and operator specialisations; experimental procedure details; statistical analysis details; deployment considerations; detailed threats-to-validity mitigation; the full SMS-to-MS degeneration proof; the adjoint extension arm; and a result-level real-defect evidence summary) are provided as separate online supplementary material.

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CRediT authorship contribution statement

Meng Li: Conceptualization, Methodology, Software, Writing, original draft. **Xiaohua Yang**: Supervision, Formal analysis, Writing, review and editing. **Jie Liu**: Investigation, Validation. **Shiyu Yan**: Data curation, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Generative AI Disclosure

The authors used AI tools under human direction for editorial proofreading, LaTeX maintenance, and consistency checks. Beyond this editorial support, Study 2's experimental data were generated by AI: the mutant generation (774 valid responses), the blinded review verdicts (747), and the third-instance arbitration were produced by Claude-family agent instances under the registered packet protocol (`PREREGISTRATION_STUDY2_v1.1`, Section 5b), with deterministic offline admission and scoring machinery applied by the authors. Study 1's data provenance is unchanged: its mutants derive from the disclosed multi-LLM cross-source campaign. All research claims, methods, statistical analyses, and conclusions in both studies were authored, verified, and approved by the human authors.

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